

The Morphosyntactic–Semantic Unity of the Early Middle Japanese Realis Form *Izenkei*: the *-ru* $\approx$ *-re* allomorphy and the so-called *kakari-musubi*¹

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Abstract

This study proposes a unified morphosyntactic–semantic account of the suffixal form *-re* found in the Early Middle Japanese realis paradigm — the so-called *izenkei* — and related phenomena. It argues that *-re* is a contextual allomorph of the suffix *-ru*, which marks a clause as [+ACT(UAL-WORLD-COMMITMENT)] and thereby presents the eventuality expressed by that clause as belonging to the actual world. It surfaces as *-re* when the clause is set as a premise.

This premise-setting environment is supplied overtly by the suffixes *-ba* [+COND(ITIONAL)] and *-do(mo)* [+CONCESSIVE], and covertly by the phonologically null suffix $\emptyset$ [+PREM(ISE-SETTING)] in the *=koso* type of *kakari-musubi*. Thus, *=koso* does not itself condition the allomorphy *-ru* $\rightarrow$ *-re*, but merely focalises a constituent not containing the finite predicate within a clause already set as a premise by $\emptyset$ [+PREM]. In the *=zo/=namu* type, by contrast, a topic–comment clause is first constructed, and a topic constituent containing the finite predicate is nominalised by $\emptyset$ [+NOM(INAL)] so that it can be treated as a nominal topic phrase. The clause is then set as a premise by $\emptyset$ [+PREM], after which *=zo* or *=namu* focalises the comment phrase. However, the intervening $\emptyset$ [+NOM] following *-ru* blocks the local morphological relation between *-ru* and $\emptyset$ [+PREM], and *-ru* therefore remains unchanged.

Accordingly, *=koso* occurs with *-re*, whereas *=zo* and *=namu* occur with *-ru*, simply because the two relevant constructions have different morphosyntactic and information-structural configurations. The particles do not determine the predicate forms. Rather, the configuration is constructed FIRST and THEN determines both the predicate form and which focus particle may occur. Traditional particle–predicate concord called *kakari-musubi* is therefore reanalysed as a surface correspondence resulting from their occurrence within the same structure, constructed independently of the so-called concord.

This study further discusses the structural parallelism between verbal and verbal-adjectival realis forms that supports the same underlying *-ru* in both categories. The analysis also accounts for the [+CONDITIONAL]-based opposition between the realis forms *-re-ba* [OPAST] and *-sika-ba* [+PAST] and the irrealis forms $\emptyset$ -*ba* [OPAST] and *-se-ba* [+PAST]. Then, the loss of productive $\emptyset$ -*ba*, and hence of the opposition between *-re-ba* [+ACT, +COND] and $\emptyset$ -*ba* [–ACT, +COND], is proposed to have caused *-re* to

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lose its [+ACT] value and to be reanalysed with *-ba* as the single conditional suffix *-reba* [~~+ACT~~, +COND].

Finally, the study then demonstrates that, in terms of the four semantic natural classes — the realis set {-*ru* [+ACT]}, the irrealis set {- $\emptyset$ [-ACT], *-zu*, *-mu*, *-zi*, *-masi*, *-mahosi*-, *-yo*}, the other irrealis set {-*ramu*, *-rasi*, *-mer*-, *-nar*-, *-be*-, *-mazi*-} and the premise-setting set {- $\emptyset$ [+PREM], *-ba*, *-do(mo)*} — and eight local allomorphies, the domains traditionally treated separately — the =*koso* type of kakari-musubi, conditional expressions and the realis–irrealis opposition — can be reduced to a single system.

1. Introduction

This section provides an overview of the principal analytical devices employed in this article. The present study argues that the suffixal form *-re* found in the Early Middle Japanese (EMJ) realis paradigm — the so-called *izenkei*, as in *m-i-re-ba* 'since/when one sees', *m-i-re-do(mo)* 'though one sees' and the construction $[[\dots =koso] [\dots m-i-re]]$ 'one sees ... exactly ...', from the verb *miru* 'look at; see' — is not an independent inflexional morpheme, but an allomorph of the suffix *-ru*, specified as [+ACT(UAL-WORLD-COMMITMENT), 0PAST], selected in premise-setting contexts. The morphosyntactic–semantic environments conditioning the allomorphy *-ru* → *-re* can be schematised as in (1)–(3), where '=' indicates enclisis:

(1) $[\dots -ru] + -\emptyset$ [+PREMISE-SETTING]

→ $[[\dots -ru]-\emptyset]_{\text{PREMISE}}$

→ $[[\dots -re]-\emptyset]_{\text{PREMISE}}$

→ $[[[\dots =koso]_{\text{FOCUS}} + [\dots -re]_{\text{BACKGROUND}}]-\emptyset]_{\text{PREMISE}}$

(2) $[\dots -ru] + -ba$ [+CONDITIONAL]

→ $[[\dots -ru]-ba]_{\text{CONDITIONAL_PROTASIS/PREMISE}}$

→ $[[\dots -re]-ba]_{\text{CONDITIONAL_PROTASIS/PREMISE}}$

→ $[[\dots -re]-ba]_{\text{CONDITIONAL_PROTASIS/PREMISE}} + [\dots]_{\text{APODOSIS}}$

(3) $[\dots -ru] + -do(mo)$ [+CONCESSIVE]

→ $[[\dots -ru]-do(mo)]_{\text{CONCESSIVE_PROTASIS/PREMISE}}$

→ $[[\dots -re]-do(mo)]_{\text{CONCESSIVE_PROTASIS/PREMISE}}$

→ $[[\dots -re]-do(mo)]_{\text{CONCESSIVE_PROTASIS/PREMISE}} + [\dots]_{\text{APODOSIS}}$

In (1), the clause comprises a potential focus and its background, while the suffix *-ru* marks that clause as [+ACT] and thereby presents the eventuality expressed by that clause as belonging to the actual world. (The [0PAST] value of *-ru* [+ACT, 0PAST], surfacing as *-re*, will be addressed in Section 6 through the opposition between [0PAST] and [+PAST] in *-re-ba* and *-\emptyset-ba*, on the one hand, and *-sika-ba* and *-se-ba*, on the other, as in *m-i-re-ba* 'since/when one sees' and *m-i-\emptyset-ba* 'if one sees' vs. *m-i-sika-ba* 'since/when one saw' and *m-i-se-ba* 'if one saw' from the aforementioned verb *miru*.) The phonologically null morpheme *-\emptyset* immediately following

-ru, specified as [+PREM(ISE-SETTING)], then sets that clause as the premise for the subsequent focalising operation, whereby *=koso* focalises the constituent corresponding to the potential focus within that clause. In (2), the suffix *-ba*, specified as [+COND(ITIONAL)], sets as a premise the clause marked as [+ACT] by *-ru* and presents the eventuality expressed by the apodosis as following naturally — in the light of background knowledge comprising general knowledge, the speech situation and the preceding context — from the eventuality expressed by the protasis. In (3), the suffix *-do(mo)*, specified as [+CONC(ESSIVE)], also sets as a premise the clause marked as [+ACT] by *-ru*, but presents the eventuality expressed by the apodosis as constituting a consequence contrary to what would ordinarily be expected from that expressed by the protasis. The common conditioning environment can therefore be formulated more simply as in (4):

$$(4) -ru [+ACT, 0PAST] \rightarrow -re / \text{PREMISE-SETTING ENVIRONMENT: } \{-\emptyset, -ba, -do(mo)\}$$

In other words, *-re* itself does not mean 'condition', 'concession' or 'background'. Rather, the allomorphy *-ru* $\rightarrow$ *-re* is triggered when a clause marked as [+ACT] by *-ru* is incorporated as a premise into another semantic structure: a focus–background structure in the case of $-\emptyset$ [+PREM] with *=koso*, a conditional structure in the case of *-ba* and a concessive structure in the case of *-do(mo)*.

Accordingly, with regard to the $[[\dots=koso] + [\dots -re]]$ type of the so-called *kakari-musubi*, this analysis does not treat *=koso* as selecting the so-called *izenkei* in *-re*, as traditionally assumed. Rather, this allomorphy occurs because the phonologically null morpheme $-\emptyset$ following *-ru*, specified as [+PREM], sets as a premise the clause marked as [+ACT] by *-ru*, thereby providing the basis for the subsequent focalisation. The particle *=koso* THEN focuses a particular constituent within that premised clause. As will be discussed in Section 4, the independently attested conditional use of bare realis forms, without *=koso* or an overt connective such as *-ba* or *-do(mo)*, indicates that *=koso* cannot be the sole or defining trigger of *-re* (cf. Ōno et al., 1990: 1497, 1501). Rather, *=koso* participates in one particular premise-setting environment in which *-ru* has been realised as *-re*.

By contrast, in the $[[\dots =zo/=namu] + [\dots -ru]]$ type of *kakari-musubi*, which ends in the adnominal form of the finite predicate, a clause containing a topic–comment information structure is first constructed. If the topic constituent contains the finite predicate, it is then nominalised by the phonologically null suffix $-\emptyset$, specified as [+NOM(INAL)], which corresponds to the Contemporary Japanese (CJ) formal noun *=no* [+NOM], so that it can be treated as a nominal topic phrase. The clause so formed is then set as a premise by $-\emptyset$ [+PREM], after which *=zo* or *=namu* focalises the comment phrase on that nominalised topic phrase, thereby yielding the focus–background structure, as represented in (5):

$$(5) [[[[\dots]_{\text{COMMENT}}=zo/=namu]_{\text{FOCUS}} + [[[\dots -ru]-\emptyset]_{\text{NOMINALISED_TOPIC}}]_{\text{BACKGROUND}}]-\emptyset]_{\text{PREMISE}}$$

Thus, the morphosyntactic and information-structural configurations are different. In the *=koso* type of *kakari-musubi*, the derivation proceeds in the order 'clause construction → premise-setting → internal focus-background structuring by *=koso*'. In the *=zo/=namu* type, by contrast, the derivation proceeds in the more complex order 'topic-comment structure → clause construction → topic-phrase nominalisation → premise-setting → internal focus-background structuring through the addition of *=zo/=namu* to the comment phrase'. Even if the clause whose topic-expressing constituent has been nominalised is set as a premise by $-\emptyset$ [+PREM] before the focalising operation, the intervening $-\emptyset$ [+NOM] immediately following *-ru* prevents $-\emptyset$ [+PREM] from standing in the local morphological relation with *-ru* that is required to trigger the allomorphy *-ru* → *-re*. Consequently, *-ru* is not realised as *-re*.

Finally, whereas the underlying form (UF) *-ru-ba* of the realis conditional form in *-re-ba*, as in *m-i-ru-ba*, surfacing as *m-i-re-ba* 'since/when one sees', contains the suffix *-ru* specified as [+ACT], the UF $-\emptyset$ -*ba* of the irrealis conditional form in *-ba*, as in *m-i-\emptyset-ba*, surfacing as *m-i-ba* 'if one sees', contains $-\emptyset$, **specified as [-ACT]**. The two forms therefore constitute a semantic opposition. Furthermore, there is independent morphophonological evidence for this phonologically null suffix $-\emptyset$, as will be discussed in Section 3. Since $-\emptyset$ in the latter construction, $-\emptyset$ -*ba*, marks the protasis as [-ACT] through its formal opposition to *-ru*, specified as [+ACT], and thereby presents the eventuality expressed by the protasis as not yet belonging to the actual world, the following *-ba*, specified as [+COND], sets the actualisation of that eventuality itself as a hypothetical condition for the eventuality expressed by the apodosis.

The essential analytical devices employed in this account are as follows:

- **Functional features:** [ACT], [PREM], [COND], [CONC], [NOM] and [PAST]
- **Semantic natural classes:** realis, irrealis and premise-setting
- **Information-structural configurations:** FOCUS + BACKGROUND and TOPIC + COMMENT
- **Morphosyntactic structure:** Radical + thematic suffix + inflexional suffix(es)
- **Morphophonological alternations**

2. The realis form *-re* as an allomorph of the suffix *-ru*

The analysis below is based on the examples in (6) and (7), where LOC, POST, POSS, GEN, THEM, ACC, V.TH, CONJ, MNR, ESS, ADD, NEG, and FOC stand for *locative*, *postposition*, *possessor*, *genitive*, *theme*, *accusative*, *verbal thematic suffix*, *conjunctive*, *manner*, *essive*, *additive*, *negative* and *focus*, respectively:

- (6) *ike-midu=ni* *koyohi=no* *tuki=wo* *utusi-mo-Ø-te*
pond-water(LOC)=POST tonight(POSS)=GEN moon(THEM)=ACC reflect-hold-V.TH-CONJ
- kokoro=no* *mama=ni* *wa=ga* *mono=to* ***m-i-ru***
mind(POSS)=GEN accordance(MNR)=POST 1sg(POSS)=GEN thing(ESS)=POST see-V.TH-ACT

'By reflecting tonight's moon in the pond water, I see it, at will, as my own.'
(Retired Emperor Shirakawa, *Kin'yō wakashū* III: 180; Kawamura et al., 2023)

- (7) *sira-yuki=no* *tokoro=mo* *wak-a-zu* *huri-sik-Ø-e-ba*
white-snow(THEM)=GEN place(THEM)=ADD distinguish-V.TH-NEG fall-spread-V.TH-ACT-COND
- ihaho=ni=mo* *sak-Ø-u* ***hana=to=koso*** ***m-i-re***
crag(LOC)=POST=ADD bloom-V.TH-ACT flower(ESS)=POST=FOC see-V.TH-ACT

'Since the white snow has fallen thick without distinguishing one place from another,
I see it as flowers that bloom even on crags — as flowers, precisely.'
(Ki no Akimine, *Kokin wakashū* VI: 324; Saeki, 1981)

The term V.TH used in (6) and (7) — namely, verbal thematic suffix — requires some explanation. Traditionally, inflected forms of verbal adjectives (VAs) in CJ, such as the conjunctive form *takaku(-)* — the so-called *ren'yōkei* — from *takai* 'be high/tall', which descends from EMJ *takasi* 'idem', have been analysed as consisting of a so-called stem such as *taka-* and the so-called ending such as *-ku*. However, based on multiple morphosyntactic–semantic parallelisms with verbal stems (V stems), such as *tabe(-)* from the verb *taberu* 'eat', descending from EMJ *tabu* 'eat/drink' (humble), Makino (2025) argues that *takaku(-)* as a whole functions as the stem of the verbal adjective (VA stem). As for the pseudo-stem *taka-* and the pseudo-ending *-ku*, the study analyses *taka-* as one of the two immediate constituents of the VA stem *takaku(-)*, which encodes, as a VA radical, the lexical meaning specific to the VA, and *-ku* as the other constituent, functioning as the VA thematic suffix — namely, the VA stem-forming suffix — which licences the VA radical as a syntactic constituent of the VA category at the sentential level. Thus, the VA category exhibits the **basic structure 'radical + thematic suffix + inflexional suffix(es)'**, which also applies to VA forms in EMJ. The study further argues that **inflected forms of true verbs in CJ and EMJ show the same basic structure.**

Building on that study, the morphosyntactic structure of the UFs of the monograde vowel-stem verb *miru* 'look at; see', the monograde vowel-stem verb *keru* 'kick' and the bigrade consonant-stem verb *kiku* 'listen to; hear' in EMJ can be analysed as illustrated in (8), where V stem, V radical and V thematic suffix stand for *verbal stem*, *verbal radical* and *verbal thematic suffix*, respectively:

- (8)

V stem		Ending
V radical	V thematic suffix	Inflexional suffix
<i>m-</i>	<i>-i</i>	<i>-ru</i>

<i>k-</i>	<i>-e</i>	<i>-ru</i>
<i>kik-</i>	<i>-Ø</i>	<i>-ru</i>

In the UF *kik-Ø-ru*, *-Ø* is a phonologically null V thematic suffix. In *kik-Ø-ru*, the suffix-initial consonant *r-* is deleted, yielding the surface form (SF) *kik-u*, owing to the surface-illicit consonant cluster *-kr-*. The alternations of the V thematic suffixes *-i* $\approx$ *-u* and *-e* $\approx$ *-u* in bigrade vowel-stem verbs (the so-called *nidan dōsi*) and the trigrade vowel-stem verbs, traditionally referred to as the *s*-irregular and *k*-irregular verbs (the so-called *sahen dōsi* and *kahen dōsi*), will be treated below. The alternation *-Ø* $\approx$ *-a* in bigrade consonant-stem verbs (the so-called *yodan dōsi*) and the trigrade consonant-stem verbs, traditionally referred to as the *n*-irregular verbs (the so-called *nahen dōsi*), and the alternations *-i* $\approx$ *-e* and *-i* $\approx$ *-o* in the trigrade vowel-stem verbs will be addressed in Section 3.

Returning to the example sentences in (6) and (7), they exhibit an alternation between [... *Vstem-ru*] and [... *=koso*] [... *Vstem-re*], as schematised in (9), where *Vstem* stands for a verbal stem consisting of a V radical followed by a V thematic suffix, such as *m-i(-)*:

- (9) *A=wo B=to m-i-ru* '(lit.) see A as B'
A=wo B=to=koso m-i-re '(lit.) see A precisely as B'

The two alternant forms, *-ru* and *-re*, occur after the same V stem *m-i-* and bear the same modal function of presenting the eventuality expressed by the clause as belonging to the actual world. The form *-ru* occurs in the unmarked construction, whereas *-re* is contextually restricted and is correlated with — but, as will be discussed in Section 4, not directly conditioned by — the presence of a phrase marked by the focus particle *=koso*. This complementary distribution supports the analysis that *-re* is a contextually conditioned allomorph of the suffix *-ru*, specified as [+ACTUAL-WORLD-COMMITMENT] or, for short, [+ACT]. The morphosyntactic–semantic environment conditioning this allomorphy will be identified in Section 4. Furthermore, the existence of the construction [... *=koso*] [... *Vstem-re*], in which *-re* is followed by neither *-ba* nor *-do(mo)*, demonstrates that *-re* cannot be analysed as an inseparable part of *-reba* or *-redo(mo)*. These forms must instead be segmented as *-re-ba* and *-re-do(mo)*, respectively, with *-re* bearing the [+ACT] value and therefore constituting a contextual allomorph of the suffix *-ru*, while *-ba* and *-do(mo)* constitute independent connective suffixes.

Inoue (2002) analyses the unmarked sentence-final predicate form in CJ as expressing 'assertion' (*kakugen*) or 'categorical assertion' (*dangen*). In his analysis, this modality is carried by a zero form *-φ* following the predicate, rather than by *-ru* itself, as he notes: '... the relevant forms each have a basic meaning: "predicate + φ" expresses ⟨assertion (categorical assertion)⟩, "*-darō*" expresses ⟨judgement formation⟩ and "*-(y)ō*" expresses ⟨intention formation⟩ ...' (Inoue 2002: 135, translated from Japanese). The present study agrees with this analysis as an account of CJ. In contrast to EMJ, virtually all modal

forms in CJ came to follow the *-ru* form, with the volitional suffix *-jō* and the imperative suffixes being the principal exceptions. In this system, *-ru* came to express [–PAST] through its paradigmatic opposition to underlying *-tar* [+PAST], which surfaces as *-ta* in most contexts. Accordingly, in the CJ structure *Vstem-ru-φ*, it is *-φ*, rather than *-ru*, that carries assertive modality. The claim of the present study is that the value carried by *-φ* in this CJ structure had instead been carried by *-ru* itself in EMJ. It should be noted that there is a clear precedent for attributing a substantive modal function to the suffix *-ru* itself in Tamura (2009), but Tamura's analysis concerns CJ rather than EMJ, and the modal value attributed to *-ru* is foreseeable futurity, which differs from the EMJ [+ACT] value proposed in the present study. Sano (2022: 7), treating Old Japanese, characterises the realis–irrealis opposition as follows: '[*p-eba q*] "*q* whenever *p*" ("*q* always when *p*")' and '[*p-aba q*] "*q* at any circumstance in which *p*". However, she assigns these meanings directly to the morphologically unanalysed subordinators *eba* and *aba*, rather than decomposing the SFs *-eba* and *-aba* into the UFs *-Ø-ru-ba* and *-Ø-Ø-ba*, respectively, and attributing [+ACT] and [–ACT] to *-ru* and *-Ø*. This is clear from her treatment of the two subordinators: 'I provide the following translation for the provisional subordinator *eba*' (Sano 2022: 7); 'The following translation can be given to the conditional subordinator *aba*' (Sano 2022: 8). The novelty of the present study, then, lies in localising in the single EMJ suffix *-ru* the semantic value [+ACT], which unifies several notions previously treated separately: actual-world commitment, assertion or categorical assertion and the realis character of the so-called *izenkei*. By doing so, the analysis provides a unified account of the conditions under which the allomorphy *-ru* → *-re* applies, the underlying opposition between *-ru-ba* and *-Ø-ba*, the phenomenon traditionally called *kakari-musubi* and the historical development from EMJ *-re-ba* to CJ *-reba*.

Returning to the analysis of the suffixal form *-re* as a contextual allomorph of the suffix *-ru*, it is supported by independent morphophonological evidence. In bigrade vowel-stem verbs, whose V thematic suffix alternates as *-i* ≈ *-u* or *-e* ≈ *-u*, only the V thematic suffixes of the conclusive, adnominal and realis forms exhibit the form *-u*, as in *nob-i* ≈ *nob-i-zu* ≈ *nob-i-te* ≈ *nob-u-Ø##* ≈ *nob-u-ru#NP* ≈ *nob-u-re-ba* ≈ *nob-i-yo* from the non-causative verb *nobu* 'stretch/extend' and *nob-e* ≈ *nob-e-zu* ≈ *nob-e-te* ≈ *nob-u-Ø##* ≈ *nob-u-ru#NP* ≈ *nob-u-re-ba* ≈ *nob-e-yo* from the causative verb *nobu* 'stretch/extend', where ## and # designate a main-clause-final boundary and a morphological word boundary that coincides with a prosodic word boundary, respectively. If the form *-re* in the realis forms *nob-u-re-ba* and *nob-u-re-do(mo)* is, as stated above, a contextual allomorph of the suffix *-ru*, the relevant UFs can be assumed to be *nob-i* ≈ *nob-i-zu* ≈ *nob-i-te* ≈ *nob-i-ru##* ≈ *nob-i-ru#NP* ≈ *nob-i-ru-ba* ≈ *nob-i-yo* and *nob-e* ≈ *nob-e-zu* ≈ *nob-e-te* ≈ *nob-e-ru##* ≈ *nob-e-ru#NP* ≈ *nob-e-ru-ba* ≈ *nob-e-yo*, respectively. Thus, the derivation of the SFs of these two verbs from their UFs can be schematised as in (10), where IF stands for *intermediate form*:

(10) UF	IF	SF
<i>nob- -i</i>	→ <i>nob- -i</i>	→ <i>nob-i</i>
<i>nob- -i -zu</i>	→ <i>nob- -i -zu</i>	→ <i>nob-i-zu</i>
<i>nob- -i -te</i>	→ <i>nob- -i -te</i>	→ <i>nob-i-te</i>
<i>nob- -i -ru ##</i>	→ <i>nob- -u -ru ##</i>	→ <i>nob-u-Ø##</i>
<i>nob- -i -ru #NP</i>	→ <i>nob- -u -ru #NP</i>	→ <i>nob-u-ru#NP</i>
<i>nob- -i -ru -ba</i>	→ <i>nob- -u -ru -ba</i>	→ <i>nob-u-re-ba</i>
<i>nob- -i -ru -do(mo)</i>	→ <i>nob- -u -ru -do(mo)</i>	→ <i>nob-u-re-do(mo)</i>
<i>nob- -i yo</i>	→ <i>nob- -i yo</i>	→ <i>nob-i-yo</i>
<i>nob- -e</i>	→ <i>nob- -e</i>	→ <i>nob-e</i>
<i>nob- -e -zu</i>	→ <i>nob- -e -zu</i>	→ <i>nob-e-zu</i>
<i>nob- -e -te</i>	→ <i>nob- -e -te</i>	→ <i>nob-e-te</i>
<i>nob- -e -ru ##</i>	→ <i>nob- -u -ru ##</i>	→ <i>nob-u-Ø##</i>
<i>nob- -e -ru #NP</i>	→ <i>nob- -u -ru #NP</i>	→ <i>nob-u-ru#NP</i>
<i>nob- -e -ru -ba</i>	→ <i>nob- -u -ru -ba</i>	→ <i>nob-u-re-ba</i>
<i>nob- -e -ru -do(mo)</i>	→ <i>nob- -u -ru -do(mo)</i>	→ <i>nob-u-re-do(mo)</i>
<i>nob- -e yo</i>	→ <i>nob- -e yo</i>	→ <i>nob-e-yo</i>

The first allomorphy of the V thematic suffix *-i* → *-u* / *-ru* in (10), found in bigrade vowel-stem verbs, is also observed in the **trigrade vowel-stem verbs** *Su* 'do' and *Ku* 'come', traditionally referred to as the *s*-irregular and *k*-irregular verbs, respectively — the so-called *sahen dōsi* and *kahen dōsi* — as demonstrated in (11), where, in order to show the default form of the thematic suffix, the V stem consisting only of the V radical and the V thematic suffix is placed at the beginning of each set of inflected forms²:

(11) UF	IF	SF
<i>S- -i</i>	→ <i>S- -i</i>	→ <i>S-i</i>
<i>S- -i -ru ##</i>	→ <i>S- -u -ru ##</i>	→ <i>S-u-Ø##</i>
<i>S- -i -ru #NP</i>	→ <i>S- -u -ru #NP</i>	→ <i>S-u-ru#NP</i>
<i>S- -i -ru -ba</i>	→ <i>S- -u -ru -ba</i>	→ <i>S-u-re-ba</i>
<i>S- -i -ru -do(mo)</i>	→ <i>S- -u -ru -do(mo)</i>	→ <i>S-u-re-do(mo)</i>
<i>K- -i #</i>	→ <i>K- -i #</i>	→ <i>K-i#</i>
<i>K- -i -ru ##</i>	→ <i>K- -u -ru ##</i>	→ <i>K-u-Ø##</i>
<i>K- -i -ru #NP</i>	→ <i>K- -u -ru #NP</i>	→ <i>K-u-ru#NP</i>
<i>K- -i -ru -ba</i>	→ <i>K- -u -ru -ba</i>	→ <i>K-u-re-ba</i>
<i>K- -i -ru -do(mo)</i>	→ <i>K- -u -ru -do(mo)</i>	→ <i>K-u-re-do(mo)</i>

Furthermore, in the trigrade consonant-stem verbs *sinu* 'die' and *inu* 'go away; leave', traditionally referred to as the *n*-irregular verbs (the so-called *nahen dōsi*), an allomorphy of the same type, *-Ø* → *-u*, also

² In order to distinguish the trigrade vowel-stem verb *ku* 'come' from the regular vowel-stem verb *kiru* 'wear', whose radical is also /k-/, the former radical /k-/ will be notated as capital K-. Similarly, the radical of the trigrade vowel-stem verb *su* 'do' will be designated as capital S-.

occurs, as indicated in (12) with *sinu*:

(12)	UF			IF				SF
	<i>sin-</i>	-Ø		<i>sin-</i>	-Ø			<i>sin</i> ³
	<i>sin-</i>	-Ø	<i>-ru</i> ##	→	<i>sin-</i>	-u	<i>-ru</i> ##	→ <i>sin-u-Ø##</i>
	<i>sin-</i>	-Ø	<i>-ru</i> #NP	→	<i>sin-</i>	-u	<i>-ru</i> #NP	→ <i>sin-u-ru#NP</i>
	<i>sin-</i>	-Ø	<i>-ru</i> <i>-ba</i>	→	<i>sin-</i>	-u	<i>-ru</i> <i>-ba</i>	→ <i>sin-u-re-ba</i>
	<i>sin-</i>	-Ø	<i>-ru</i> <i>-do(mo)</i>	→	<i>sin-</i>	-u	<i>-ru</i> <i>-do(mo)</i>	→ <i>sin-u-re-do(mo)</i>

A further consequence of the **radical-thematic_suffix-ending model** is that the traditional *yodan* and *nahen* verb classes can be reorganised as bigrade and trigrade consonant-stem verbs, respectively. In *yodan* verbs, the phonologically null V thematic suffix *-Ø* alternates with *-a*, as will be discussed in Section 3, whereas in *nahen* verbs it alternates with both *-a* and *-u*. The vowel *-i-* in forms such as *sin-i-te* is excluded from this alternation, since it is analysed as epenthetic. This reclassification is not available under the traditional stem-ending model, since it crucially depends on the independent postulation of a V thematic suffix between the radical and the inflexional ending.

The alternation *-i* → *-u* or *-e* → *-u* in the V thematic suffix found in bigrade vowel-stem verbs and trigrade vowel-stem verbs, traditionally called the *s*-irregular and *k*-irregular verbs, and the alternation *-Ø* → *-u* in the V thematic suffix found in trigrade consonant-stem verbs, traditionally called the *n*-irregular verbs, can then be analysed as occurring only immediately before underlying *-ru*. This makes it possible to describe these local alternations in an extremely simple way, as *-i/-e* → *-u* / *_-ru* and *-Ø* → *-u* / *_-ru*. The former type of allomorphy does not take place, however, in monograde vowel-stem verbs, the so-called *itidan dōsi*, such as the verbs *miru* and *keru* cited above, as indicated in (13):

(13)	UF			IF				SF
	<i>m-</i>	-i		→	<i>m-</i>	-i		→ <i>m-i</i>
	<i>m-</i>	-i	<i>-ru</i> ##	→	<i>m-</i>	-i	<i>-ru</i> ##	→ <i>m-i-ru##</i>
	<i>m-</i>	-i	<i>-ru</i> #NP	→	<i>m-</i>	-i	<i>-ru</i> #NP	→ <i>m-i-ru#NP</i>
	<i>m-</i>	-i	<i>-ru</i> <i>-ba</i>	→	<i>m-</i>	-i	<i>-ru</i> <i>-ba</i>	→ <i>m-i-re-ba</i>
	<i>m-</i>	-i	<i>-ru</i> <i>-do(mo)</i>	→	<i>m-</i>	-i	<i>-ru</i> <i>-do(mo)</i>	→ <i>m-i-re-do(mo)</i>
	<i>k-</i>	-e		→	<i>k-</i>	-e		→ <i>k-e</i>
	<i>k-</i>	-e	<i>-ru</i> ##	→	<i>k-</i>	-e	<i>-ru</i> ##	→ <i>k-e-ru##</i>

³ The final vowel *-i* in the SF *sin-i##*, derived from the UF *sin-Ø##*, is analysed as epenthetically inserted to resolve the surface-illicit word-final consonant *n* in the Yamato lexical stratum. The same *i* is epenthesised in the derivation UF *sin-Ø-te* → IF *sin-te* → SF *sin-i-te* 'die and', in order to resolve the illicit consonant cluster *-nt-* in that stratum — namely, a nasal followed by a voiceless plosive. This *i* does not occur in the alternative derivation UF *sin-Ø-te* → IF *sin-te* → SF *sin-de*, triggered by postnasal voicing. The fact that *i*-epenthesis and postnasal voicing occur in mutual exclusion demonstrates that the insertion of *i* serves to resolve *-nt-*. In UF *sin-Ø##* → IF *sin##* → SF *sin-i##*, on the other hand, no alternative repair strategy is available, so *i* must be inserted, since deletion of the illicit word-final *n* would impair the identification of the morpheme *sin-*.

<i>k-</i>	<i>-e</i>	<i>-ru</i>	<i>#NP</i>	→	<i>k-</i>	<i>-e</i>	<i>-ru</i>	<i>#NP</i>	→	<i>k-e-ru#NP</i>
<i>k-</i>	<i>-e</i>	<i>-ru</i>	<i>-ba</i>	→	<i>k-</i>	<i>-e</i>	<i>-ru</i>	<i>-ba</i>	→	<i>k-e-re-ba</i>
<i>k-</i>	<i>-e</i>	<i>-ru</i>	<i>-do(mo)</i>	→	<i>k-</i>	<i>-e</i>	<i>-ru</i>	<i>-do(mo)</i>	→	<i>k-e-re-do(mo)</i>

Accordingly, the allomorphies of the V thematic suffixes *-i*, *-e* and *-Ø* in (10)–(12) can be formulated as in (14) and (15), where the term *multigrade vowel-stem verbs* includes bigrade vowel-stem verbs and trigrade vowel-stem verbs:

(14) V thematic suffixes *-i* and *-e* → *-u* / *_-ru* in multigrade vowel-stem verbs

(15) V thematic suffix *-Ø* → *-u* / *_-ru* in trigrade consonant-stem verbs

In (10) – (12), in addition to the allomorphy *-ru* → *-re* / *_-ba, -do(mo)*, another allomorphy, *-ru* → *-Ø* / *_##*, is postulated. Since SFs such as *nob-u-re-ba/-do(mo)*, *nob-u-Ø##* and *nob-u-ru#NP* are all specified as [+ACT], the complementarily distributed forms *-re*, *-Ø* and *-ru* are assumed to be three surface exponents of the same [+ACT] morpheme. This latter allomorphy, *-ru* → *-Ø* / *_##*, does not occur, however, in bigrade consonant-stem verbs other than *r*-irregular verbs such as *ari* 'exist' — the so-called *rahen dōsi* — or in monograde vowel-stem verbs, and is therefore restricted as formulated in (16):

(16) *-ru* → *-Ø* / *_##* in trigrade consonant-stem verbs, multigrade vowel-stem verbs and *r*-irregular verbs

3. Semantic opposition between the realis form *izenkei* and the irrealis form *mizenkei* before the conditional suffix *-ba*

In parallel with the interpretation that the form *-re* in the construction [=koso ... *Vstem-re*], as in *hana=to=koso m-i-re* 'see precisely as flower' in (7), is a contextual allomorph of the suffix *-ru* found in *Vstem-ru*, as in *wa=ga mono=to m-i-ru* 'see as my own' in (6), the form *-re* in *Vstem-re-ba*, as in *m-i-re-ba* 'since/when one sees' from the verb *miru* 'look at; see', is likewise assumed to be the same contextual allomorph of the suffix *-ru* — the allomorphy being conditioned, in this case, by the following conditional suffix *-ba*.

The SF *Vstem-re-ba*, which thus corresponds to the UF *Vstem-ru-ba*, is traditionally referred to as 'realis form (*izenkei*) + particle (*zyosi*) *-ba*' and is semantically distinct from the SF *Vstem-ba*, such as *m-i-ba* 'if one sees', which corresponds to the UF *Vstem-Ø-ba* and is traditionally referred to as 'irrealis form (*mizenkei*) + particle (*zyosi*) *-ba*', as shown in (17) and (18), where PFV, EP.NEC, NA.TH, RES, APPOS and SCOPLIM stand for *perfective*, *epistemic necessity*, *nominal-adjectival thematic suffix*, *resultative*, *apposition* and *scope-limiting*, respectively:

- (17) *ori-Ø-te* *m-i-Ø-ba* *oti-Ø=zo*
 break.off-V.TH-CONJ see-V.TH-[-ACT]-COND fall-V.TH=FOC
- s-i-n-u-be-ki-Ø-Ø* *aki-hagi=no* *eda=mo*
 do-V.TH-PFV-V.TH-EP.NEC-VA.TH-ACT-NOM autumn-bush.clover(POSS)=GEN branch(THEM)=ADD
- tawawa-ni* *ok-Ø-er-Ø-u* *sira-tuyu*
 heavily.laden-NA.TH settle-V.TH-RES-V.TH-ACT white-dew(THEM)

'If I break off [a branch] and look at it, the white dew resting so heavily on the autumn bush-clover's branches that even they bend will surely fall.'
 (Anonymous, *Kokin wakashū* IV: 223; Saeki, 1981)

- (18) *tuki* *m-i-re-ba* *tidi-ni* *mono=koso*
 moon(THEM) see-V.TH-ACT-COND manifold-NA.TH thing(THEM)=FOC
- kanasi-ke-re-Ø* *[[wa=ga mi] hitotu]=no*
 be.sad-VA.TH-ACT-PREM [[1sg(POSS)=GEN body(POSS)] one(APPOS)]=GEN
- aki=ni=wa* *ar-a-ne-do*
 autumn(LOC)=POST=SCOPE.LIM exist-V.TH-NEG-CONC

'When I look at the moon, things in countless ways fill me with sadness, although it does not exist as autumn belonging to me alone'
 (Ōe no Chisato, *Kokin wakashū* IV: 193; Saeki, 1981)

In the former construction *Vstem-re-ba* — underlyingly *Vstem-ru-ba* — the suffix *-ru*, specified as [+ACT], first marks the protasis as [+ACT] and thereby presents the eventuality expressed by the protasis as belonging to the actual world. The suffix *-ba*, specified as [+COND], then sets that protasis as a premise and presents the eventuality expressed by the apodosis as following naturally from that expressed by the protasis, in the light of background knowledge comprising general knowledge, the speech situation and the preceding context. Therefore, *-ba* itself does not lexically signify cause, occasion or general condition as distinct meanings. Rather, these interpretations are assumed to be pragmatically determined by the interaction between the protasis, the apodosis and the background knowledge.

The latter construction *Vstem-ba*, on the other hand, first presents the eventuality expressed by the protasis as not belonging to the actual world and then presents the actualisation of the eventuality itself as the condition for the establishment of the eventuality expressed by the apodosis. For example, *m-i-re-ba* means 'since/when one sees', whereas *m-i-ba* means 'if one sees'. Note that *i-zen-* in the Japanese grammatical term *i-zen-kei* literally signifies 'already (*i-*) so (*-zen-*)', whereas *mi-zen-* in *mi-zen-kei* signifies 'not yet (*mi-*) so (*-zen-*)'.

Formally, the only difference between the UFs *Vstem-ru-ba* and *Vstem-Ø-ba* is that the former contains the suffix *-ru*, whereas the latter contains *-Ø* in its place. Accordingly, the suffix *-ru* in the UF *Vstem-ru-*

ba, specified as [+ACT], marks the protasis clause as [+ACT] and thereby presents the eventuality expressed by that clause as belonging to the actual world, whereas **the phonologically null suffix $-\emptyset$ in the UF $Vstem-\emptyset-ba$, specified as [-ACT], marks the protasis clause as [-ACT] and thereby presents the eventuality expressed by that clause as not belonging to the actual world through its formal opposition to $-ru$** . This is parallel to the case of English *cat- $\emptyset$* , where the lack of an overt plural marker in opposition to *cat-s* functions to mark the singular. The structural and semantic opposition between the UFs $Vstem-ru-ba$ and $Vstem-\emptyset-ba$ may be schematised as in (19):

(19)	UF		SF	Interpretation
	$Vstem$	$-ru$ [+ACT]	$-ba$ [+COND]	$\rightarrow$ $Vstem$ $-re$ $-ba$ cause, occasion or general condition
	$Vstem$	$-\emptyset$ [-ACT]	$-ba$ [+COND]	$\rightarrow$ $Vstem$ $-ba$ hypothetical condition

This analysis receives independent morphophonological support from consonant-stem verbs. If the SFs $m-i-re-ba$ and $m-i-ba$ correspond to the UFs $m-i-ru_{[+ACT]}-ba$ and $m-i-\emptyset_{[-ACT]}-ba$, respectively, the corresponding SFs $kik-e-ba$ 'since/when one hears' and $kik-a-ba$ 'if one hears' from the bigrade consonant-stem verb $kik-u$ 'listen to; hear', and $sin-u-re-ba$ 'since/when one dies' and $sin-a-ba$ 'if one dies' from the trigrade consonant-stem verb $sin-u$ 'die', both of which contain the V thematic suffix $-\emptyset$, must likewise be derived from the UFs $kik-\emptyset-ru_{[+ACT]}-ba$, $kik-\emptyset-\emptyset_{[-ACT]}-ba$, $sin-\emptyset-ru_{[+ACT]}-ba$ and $sin-\emptyset-\emptyset_{[-ACT]}-ba$, as shown in (20):

(20)	UF				IF				SF		
	<i>kik-</i>	$-\emptyset$	-ru [+ACT]	<i>-ba</i>	$\rightarrow$	<i>kik-</i>	$-\emptyset$	-re	<i>-ba</i>	$\rightarrow$	<i>kik-e-ba</i>
	<i>kik-</i>	$-\emptyset$	$-\emptyset$ [-ACT]	<i>-ba</i>	$\rightarrow$	<i>kik-</i>	-a	$-\emptyset$	<i>-ba</i>	$\rightarrow$	<i>kik-a-ba</i>
	<i>sin-</i>	$-\emptyset$	-ru [+ACT]	<i>-ba</i>	$\rightarrow$	<i>sin-</i>	<i>-u</i>	-re	<i>-ba</i>	$\rightarrow$	<i>sin-u-re-ba</i>
	<i>sin-</i>	$-\emptyset$	$-\emptyset$ [-ACT]	<i>-ba</i>	$\rightarrow$	<i>sin-</i>	-a	$-\emptyset$	<i>-ba</i>	$\rightarrow$	<i>sin-a-ba</i>

In the derivation of the surface realis forms $kik-e-ba$ and $sin-u-re-ba$, the element immediately following the V thematic suffix $-\emptyset$ is not $-ba$, but the independent inflectional suffix $-\emptyset$ [-ACT] in opposition to $-ru$ [+ACT]. It is therefore this phonologically null suffix, rather than the connective suffix $-ba$, that locally conditions the allomorphy of V thematic suffix $-\emptyset \rightarrow -a$. The same allomorphy occurs before the negative suffix $-zu$, the assumptive suffix $-mu$, the negative-assumptive suffix $-zi$, the counterfactual suffix $-masi$ and the desiderative suffix $-mahosi$, as in $kik-\emptyset-zu \rightarrow kik$ **$-a$** $-zu$ 'one does not hear', $kik-\emptyset-mu \rightarrow kik$ **$-a$** $-mu$ 'one will hear', $kik-\emptyset-zi \rightarrow kik$ **$-a$** $-zi$ 'one will not hear', $kik-\emptyset-masi \rightarrow kik$ **$-a$** $-masi$ 'one would hear' and $kik-\emptyset-mahosi \rightarrow kik$ **$-a$** $-mahosi$ 'one wants to hear'⁴; I want to one to hear', and $sin-\emptyset-zu \rightarrow sin$

⁴ The surface conclusive form $kik-a-mahosi$ is assumed to be derived as follows: UF $kik-\emptyset-mahosi-ku-ru \rightarrow kik$ **$-a$** $-mahosi-ku-\emptyset \rightarrow kik$ **$-a$** $-mahosi$ **$-si$** $-\emptyset \rightarrow$ SF kik **$-a$** $-mahosi$ **$-\emptyset$** $-\emptyset$, with deletion of the allomorph $-si$ of the VA thematic suffix $-ku$ owing to homophonic collision with the suffix-final $-si$ of $-mahosi$ (cf. figure (39) in Section 5).

a-zu 'one does not die', *sin-Ø*-mu → *sin-a*-mu 'one will die', *sin-Ø*-zi → *sin-a*-zi 'one will not die' and *sin-Ø*-masi → *sin-a*-masi 'one would die', as illustrated in (21):

(21)	UF		SF
	<i>kik- -Ø -zu</i>	→	<i>kik- -a -zu</i>
	<i>kik- -Ø -mu</i>	→	<i>kik- -a -mu</i>
	<i>kik- -Ø -zi</i>	→	<i>kik- -a -zi</i>
	<i>kik- -Ø -masi</i>	→	<i>kik- -a -masi</i>
	<i>kik- -Ø -mahosi-</i>	→	<i>kik- -a -mahosi-</i>
	<i>sin- -Ø -zu</i>	→	<i>sin- -a -zu</i>
	<i>sin- -Ø -mu</i>	→	<i>sin- -a -mu</i>
	<i>sin- -Ø -zi</i>	→	<i>sin- -a -zi</i>
	<i>sin- -Ø -masi</i>	→	<i>sin- -a -masi</i>
	<i>sin- -Ø -mahosi-</i>	→	<i>sin- -a -mahosi-</i>

The five suffixes -Ø [-ACT], -zu, -mu, -zi, -masi form a semantic natural class in that none of them links the eventuality expressed by the verbal stem with its argument(s) to the actual world. The allomorphy can therefore be formulated as in (22):

(22) V thematic suffix -Ø → -a / _{-Ø [-ACT], -zu, -mu, -zi, -masi, -mahosi-}

Normal bigrade vowel-stem verbs such as *nobu* (non-causative) and *nobu* (causative), discussed in the last paragraph of the previous section, do not exhibit this type of allomorphy in the context _{-Ø [-ACT], -zu, -mu, -zi, -masi, -mahosi-}, as in UF *nob-i*-zu → SF *nob-i*-zu and UF *nob-e*-zu → SF *nob-e*-zu. By contrast, the thematic suffix -i in the *s*-irregular verb *su* 'do' and the *k*-irregular verb *ku* 'come' exhibits not only the allomorphy -i → -u / -ru, in the same way as the thematic suffix -i of regular bigrade vowel-stem verbs such as *nobu*, as shown in (23) (= (10) and (11)), but also the allomorphies -i → -e and -i → -o, respectively, precisely in the contexts _{-Ø [-act], -zu, -mu, -zi, -masi, -mahosi-}, and, in addition, in the context _yo, where -yo is the imperative suffix, as demonstrated in (24):

(23)	UF		IF		SF
	<i>nob- -i</i>	→	<i>nob- -i</i>	→	<i>nob-i</i>
	<i>nob- -i -ru ##</i>	→	<i>nob- -u -ru ##</i>	→	<i>nob-u-Ø##</i>
	<i>nob- -i -ru #NP</i>	→	<i>nob- -u -ru #NP</i>	→	<i>nob-u-ru#NP</i>
	<i>nob- -i -ru -Ø [+PREM]</i>	→	<i>nob- -u -ru -Ø</i>	→	<i>nob-u-re</i>
	<i>nob- -i -ru -ba</i>	→	<i>nob- -u -ru -ba</i>	→	<i>nob-u-re-ba</i>
	<i>nob- -i -ru -do(mo)</i>	→	<i>nob- -u -ru -do(mo)</i>	→	<i>nob-u-re-domo</i>
	<i>S- -i</i>	→	<i>S- -i</i>	→	<i>S-i</i>
	<i>S- -i -ru #NP</i>	→	<i>S- -u -ru #NP</i>	→	<i>S-u-ru#NP</i>
	<i>S- -i -ru -Ø [+PREM]</i>	→	<i>S- -u -ru -Ø</i>	→	<i>S-u-re</i>
	<i>S- -i -ru -ba</i>	→	<i>S- -u -ru -ba</i>	→	<i>S-u-re-ba</i>

<i>S-</i> -i <i>-ru</i> <i>-do(mo)</i>	→	<i>S-</i> -u <i>-ru</i> <i>-do(mo)</i>	→	<i>S-u-re-domo</i>
<i>K-</i> -i	→	<i>K-</i> -i	→	<i>K-i</i>
<i>K-</i> -i <i>-ru</i> ##	→	<i>K-</i> -u <i>-ru</i> ##	→	<i>K-u-Ø##</i>
<i>K-</i> -i <i>-ru</i> #NP	→	<i>K-</i> -u <i>-ru</i> #NP	→	<i>K-u-ru#NP</i>
<i>K-</i> -i <i>-ru</i> <i>-Ø</i> [+PREM]	→	<i>K-</i> -u <i>-ru</i> <i>-Ø</i>	→	<i>K-u-re</i>
<i>K-</i> -i <i>-ru</i> <i>-ba</i>	→	<i>K-</i> -u <i>-ru</i> <i>-ba</i>	→	<i>K-u-re-ba</i>
<i>K-</i> -i <i>-ru</i> <i>-do(mo)</i>	→	<i>K-</i> -u <i>-ru</i> <i>-do(mo)</i>	→	<i>K-u-re-domo</i>

(24) UF	SF
<i>S-</i> -i	→ <i>S-</i> -i
<i>S-</i> -i <i>-Ø</i> [-ACT] <i>-ba</i>	→ <i>S-</i> -e <i>-Ø</i> <i>-ba</i>
<i>S-</i> -i <i>-zu</i>	→ <i>S-</i> -e <i>-zu</i>
<i>S-</i> -i <i>-mu</i>	→ <i>S-</i> -e <i>-mu</i>
<i>S-</i> -i <i>-zi</i>	→ <i>S-</i> -e <i>-zi</i>
<i>S-</i> -i <i>-masi</i>	→ <i>S-</i> -e <i>-masi</i>
<i>S-</i> -i <i>-mahosi-</i>	→ <i>S-</i> -e <i>-mahosi-</i>
<i>S-</i> -i <i>-yo</i>	→ <i>S-</i> -e <i>-yo</i>
<i>K-</i> -i	→ <i>K-</i> -i
<i>K-</i> -i <i>-Ø</i> [-ACT] <i>-ba</i>	→ <i>K-</i> -o <i>-Ø</i> <i>-ba</i>
<i>K-</i> -i <i>-zu</i>	→ <i>K-</i> -o <i>-zu</i>
<i>K-</i> -i <i>-mu</i>	→ <i>K-</i> -o <i>-mu</i>
<i>K-</i> -i <i>-zi</i>	→ <i>K-</i> -o <i>-zi</i>
<i>K-</i> -i <i>-masi</i>	→ <i>K-</i> -o <i>-masi</i>
<i>S-</i> -i <i>-mahosi-</i>	→ <i>S-</i> -e <i>-mahosi-</i>
<i>S-</i> -i <i>-yo</i>	→ <i>S-</i> -e <i>-yo</i>

The imperative suffix *-yo* shares a semantic property with *-Ø* [-ACT], *-zu*, *-mu*, *-zi*, *-masi* and *-mahosi-* in that it presents the eventuality expressed by the clause as not yet realised and therefore as not belonging to the actual world. The alternations may thus be formulated as in (25) and (26):

(25) V thematic suffix *-i* → *-e* / *-Ø* [-ACT], *-zu*, *-mu*, *-zi*, *-masi*, *-mahosi-*, *-yo* in the *s*-irregular verb

(26) V thematic suffix *-i* → *-o* / *-Ø* [-ACT], *-zu*, *-mu*, *-zi*, *-masi*, *-mahosi-*, *-yo* in the *k*-irregular verb

The allomorphies of the V thematic suffixes *-Ø* → *-a* in consonant-stem verbs, *-i* → *-e* in the *s*-irregular verb and *-i* → *-o* in the *k*-irregular verb provide independent morphophonological evidence that *-Ø* [-ACT] in the UF *m-i-Ø-ba* forms, together with the suffixes *-zu*, *-mu*, *-zi*, *-masi*, *-mahosi-* and *-yo*, a semantic natural class, and that it is an independent morpheme standing in paradigmatic opposition to *-ru* [+ACT], rather than the mere absence of *-ru*. (Note that if the suffix used in the formation of imperative forms of consonant-stem verbs such as *kiku* were *-yo*, the derivation UF *kik-Ø-yo* would incorrectly yield SF **kik-a-yo*. The actual SF, however, is *kik-e*. It follows, therefore, that the suffix X, surfacing as *-e*, employed in the formation of imperative forms of consonant-stem verbs is distinct from *-yo* and does not trigger the

allomorphy of the thematic suffix $-\emptyset \rightarrow -a$.)

Furthermore, as will be discussed in Section 7, by recognising the modal value [+ACT] for $-ru$ and the opposite modal value [−ACT] for $-\emptyset$, both the synchronic opposition in EMJ between the realis conditional $-re-ba$ [+ACT, +COND] and the irrealis conditional $-\emptyset-ba$ [−ACT, +COND], and the diachronic reanalysis from $-re-ba$ [+ACT, +COND] to $-reba$ [~~+ACT~~, +COND], can be explained in terms of the establishment and subsequent neutralisation of one and the same opposition.

Finally, in the underlying concessive construction $Vstem-ru-do(mo)$, surfacing as $Vstem-re-do(mo)$ — as in the SF *m-i-re-domo* 'although one sees', derived from the UF *m-i-ru-domo* — the suffix $-ru$ presents the eventuality expressed by that clause as belonging to the actual world. The suffix $-do(mo)$, specified as [+CONCESSIVE] or, for short, [+CONC], then presents the eventuality expressed by the apodosis as constituting a consequence contrary to what would ordinarily be expected from that expressed by the protasis. Accordingly, $-do(mo)$ itself does not lexically distinguish factual from general concession. Rather, these interpretations are assumed to be pragmatically determined by the interaction between the protasis, the apodosis and the background knowledge. This structural and semantic analysis may be schematised as in (27):

(27)	UF		SF	Interpretation
	$Vstem$	$-ru$ [+ACT]	$-do(mo)$ [+CONC]	$\rightarrow$ $Vstem-re-do(mo)$ factual or general concession

4. The premise-setting environment created by $-ba$, $-do(mo)$ and $-\emptyset$

In the construction $[[\dots Vstem-ru]-ba]_{\text{PROTASIS}} + [\dots]_{\text{APODOSIS}}$, the protasis ending in $Vstem-ru-$, surfacing as $Vstem-re-$, constitutes the premise required for the eventuality expressed by the apodosis to stand in a causal, occasion-based or general condition–consequence relation to the eventuality expressed by the protasis, which is presented in advance as belonging to the actual world. In other words, by expressing in advance an eventuality presented as belonging to the actual world, the protasis makes it possible for the eventuality expressed by the apodosis to be placed in such a relation. Similarly, in the construction $[[\dots Vstem-ru]-do(mo)]_{\text{PROTASIS}} + [\dots]_{\text{APODOSIS}}$, the protasis ending in $Vstem-ru-$, surfacing as $Vstem-re-$, constitutes the premise required for the eventuality expressed by the apodosis to stand in a relation of factual or general concession to the eventuality expressed by the protasis, which is presented in advance as belonging to the actual world. In other words, by expressing in advance an eventuality presented as belonging to the actual world, the protasis makes it possible for the eventuality expressed by the apodosis to be placed in such a relation. Therefore, the suffixes $-ba$ and $-do(mo)$ have the function of presenting the protasis ending in $Vstem-ru-$ as a premise. The suffix $-ru$ is therefore assumed to be realised as $-re$ when the clause it marks is placed in such a premise-setting environment.

Why, then, is *-ru* realised as *-re* in the so-called *kakarimusubi* construction $[[\dots =koso] + [\dots Vstem\textbf{-ru}]]$, as in $[[\dots =koso] + [\dots Vstem\textbf{-re}]]$? In this construction, the phrase marked by the particle *=koso* constitutes the informational focus and stands in a focus–background relation to the remainder of the clause. It may therefore be represented as $[[\dots =koso]_{\text{FOCUS}} + [\dots Vstem\textbf{-re}]_{\text{BACKGROUND}}]$. For a focus–background structure to be established, the clause to which that structure applies must already have been constructed. In other words, that clause constitutes the premise for the establishment of the focus–background structure. It follows that the entire clause, underlyingly ending in *Vstem-ru*, is set as a premise before the focus–background structure is established, as represented in (28):

$$(28) [\dots Vstem\textbf{-ru}] \rightarrow [\dots Vstem\textbf{-ru}]_{\text{PREMISE}}$$

This is analogous to the requirement that, within a word, prominence realised as stress can be assigned to a particular syllable only after the syllable sequence has already been determined; focus is likewise a form of prominence assigned within an already constructed clause, as in (29):

$$(29) \text{Stress assignment: premise } import [im.pɔ:t] \rightarrow \text{focalised } ['im.pɔ:t] \text{ (noun) or } [im.'pɔ:t] \text{ (verb)}$$

Thus, in the so-called *kakarimusubi* construction of this type, the clause that the suffix *-ru* marks as [+ACT] is also placed in the same premise-setting environment, and *-ru* is therefore assumed to be realised as *-re*, as shown in

$$(30) [\dots Vstem\textbf{-ru}] \rightarrow [\dots Vstem\textbf{-ru}]_{\text{PREMISE}} \rightarrow [\dots Vstem\textbf{-re}]_{\text{PREMISE}}$$

Nevertheless, the element which marks the clause $[\dots Vstem\textbf{-ru}]$ as a premise is phonologically covert. This study therefore assumes a phonologically null morpheme $\emptyset$, specified as [+PREMISE-SETTING] or, for short, [+PREM], and posits the structure and allomorphy of *-ru* in (31):

$$(31) [\dots Vstem\textbf{-ru}] + \emptyset [+PREM] \rightarrow [[\dots Vstem\textbf{-ru}]\textbf{-}\emptyset]_{\text{PREMISE}} \rightarrow [[\dots Vstem\textbf{-re}]\textbf{-}\emptyset]_{\text{PREMISE}}$$

The independently attested conditional use of bare realis forms, without *=koso* or an overt connective such as *-ba* or *-do(mo)*, as in (32), supports this analysis, where HON stands for *honorific*:

(32)	<i>ipe-zakari-∅</i> from.home-move.away-V.TH	<i>imas-∅-u</i> go.HON-V.TH-ACT	<i>wa-g-imo=wo</i> 1sg(POSS)-GEN-beloved.woman(THEM)=ACC
	<i>todom-e-kan-e</i> restrain-V.TH-be.unable-V.TH	<i>yama</i> mountain(LOC)	<i>kakusi-∅-t-u-re-∅</i> hide-V.TH-PFV-V.TH-ACT-PREM

kokoro-do=mo *na-si-Ø*
 presence.of.mind(THEM)=ADD be.absent-VA.TH-ACT

'Unable to hold back my beloved wife as she departed from home,
 I hid her in the mountain; thus I have no presence of mind.'
 (Ōtomo no Yakamochi, *Man'yōshū* III: 471; Satake et al., 2013)

The bare *izenkei* construction indicates that premise-setting, and hence the allomorphy *-ru* → *-re*, is logically and historically prior to focalisation by *=koso*. Thus, a clause was first marked as a premise by -Ø [+PREM], with the resulting realisation of *-ru* as *-re*: *=koso* was subsequently introduced to focalise a particular constituent within that ALREADY PREMISED clause. Ōno et al. (1990: 1497, translated from Japanese) state:

"To understand diachronically how the concord between *=koso* and the *izenkei* arose, it is first necessary to recognise that, at the earliest stage, the *izenkei* was an inflectional form capable by itself of forming a conditional protasis without *=koso*. ... Thus, the *izenkei* alone could express both a non-concessive conditional protasis meaning 'since/because ...' and a concessive conditional protasis meaning 'although ...'. When *=koso* was introduced into a construction in which the *izenkei* already had this function, expressions of the following kind arose. ... In other words, the introduction of *=koso*, through its interaction with the *izenkei*, formed a concessive established-condition protasis. ... It was subsequently forgotten that *=koso* ... *izenkei* formed a concessive conditional protasis ...; ... *=koso* came to be introduced simply as an emphatic particle, while the *izenkei* came to be regarded as one of the ordinary means of concluding a sentence."

Interpreted within the framework of the present study, this development may be represented as follows:

1. The addition of -Ø [+PREM] derives a premised clause ending in the *izenkei*.
2. Depending on the context, this premised clause receives either a [+COND] or a [+CONC] interpretation.
3. *=Koso* is introduced into a premised clause receiving a [+CONC] interpretation, deriving a [+CONC] clause internally structured into focus and background.
4. The [+CONC] interpretation is subsequently lost.

Accordingly, the present study's postulation of -Ø [+PREM] and of the allomorphy *-ru* → *-re* conditioned by the premise-setting environment it provides constitutes a set of conceptual devices that succinctly accounts for the historical facts described by Ōno et al.: the historical priority of the bare *izenkei* and the

secondary introduction of *=koso*.

Based on the above discussion, the morphosyntactic–semantic environments conditioning the allomorphy *-ru* → *-re* may be represented as in (33)–(35) = (1)–(3):

(33) [... *-ru*] + *-Ø* [+PREMISE-SETTING]

→ [... *-ru*]-*Ø*_{PREMISE}

→ [... *-re*]-*Ø*_{PREMISE}

→ [[[... *=koso*]_{FOCUS} + [... *-re*]_{BACKGROUND}]-*Ø*]_{PREMISE}

(34) [... *-ru*] + *-ba* [+CONDITIONAL]

→ [... *-ru*]-*ba*_{CONDITIONAL_PROTASIS/PREMISE}

→ [... *-re*]-*ba*_{CONDITIONAL_PROTASIS/PREMISE}

→ [... *-re*]-*ba*_{CONDITIONAL_PROTASIS/PREMISE} + [...]_{APODOSIS}

(35) [... *-ru*] + *-do(mo)* [+CONCESSIVE]

→ [... *-ru*]-*do(mo)*_{CONCESSIVE_PROTASIS/PREMISE}

→ [... *-re*]-*do(mo)*_{CONCESSIVE_PROTASIS/PREMISE}

→ [... *-re*]-*do(mo)*_{CONCESSIVE_PROTASIS/PREMISE} + [...]_{APODOSIS}

The common conditioning environment can therefore be formulated more simply as in (36) = (4):

(36) *-ru* → *-re* / PREMISE-SETTING ENVIRONMENT: *-Ø*, *-ba*, *-do(mo)*

By contrast, in the [... *=zo/=namu*] + [... *-ru*] type of *kakari-musubi*, which ends in the adnominal form of the finite predicate, a clause containing a topic–comment information structure is first constructed. If the topic-expressing constituent contains a finite predicate, such as a verbal stem (*Vstem*) + the inflexional suffix *-ru*, the constituent ending in that finite predicate, as in *Vstem-ru*, is nominalised by the phonologically null suffix *-Ø*, specified as [+NOM(INAL)] — which corresponds to the CJ formal noun *=no* [+NOM] — so that it can be treated as a nominal topic phrase. The postulation of *-Ø* [+NOM] is supported by the historical observation that adnominal forms in EMJ could function nominally without the overt formal noun *=no*. The null suffix *-Ø* [+NOM] thus represents this nominalising function

morphosyntactically, with CJ =no [+NOM] as its overt counterpart, as shown in (37):

- (37) *tuki=no kaho m-i-ru-Ø_[+NOM]=wa im-Ø-u koto*
 moon(POSS)=GEN face(THEM) look.at-V.TH-ACT-NOM(THEM)=TOP avoid-V.TH-ACT thing
- 'Looking at the face of the moon (is) something to be avoided as a taboo.'
 CJ paraphrase: *tuki=no kao=wo m-i-ru=no_[+NOM]=wa im-Ø-u(-be-ki) koto(=d-a)*
 (*Taketori monogatari*, 'Kaguyahime no syōten'; Sakakura, 1970)

The clause so formed is then set as a premise for subsequent focalisation, in which the comment constitutes the focus and the topic the background. The comment phrase on that nominalised topic phrase is not marked by =*koso* but by =*zo* or =*namu*, thereby yielding the comment–topic and focus–background structure, as represented in (38):

The clause so formed is then set as a premise for subsequent focalisation, in which the comment constitutes the focus and the topic the background. The comment phrase on that nominalised topic phrase is marked not by =*koso* but by =*zo* or =*namu*, thereby yielding a structure that is at once a comment–topic one and a focus–background one, as represented in (38):

- (38) [[...]_{COMMENT} + [... *Vstem-ru*]_{TOPIC}]
 → [[...]_{COMMENT} + [[... *Vstem-ru*]_{TOPIC} + -Ø [+NOMINAL]]]
 → [[...]_{COMMENT} + [[... *Vstem-ru*]-Ø]_{NOMINALISED_TOPIC}]
 → [[...]_{COMMENT} + [[... *Vstem-ru*]-Ø]_{NOMINALISED_TOPIC}] + -Ø [+PREMISE-SETTING]
 → [[[...]_{COMMENT} + [[... *Vstem-ru*]-Ø]_{NOMINALISED_TOPIC}]-Ø]_{PREMISE}
 → [[[[...]_{COMMENT} + =*zo*/=*namu*] + [[... *Vstem-ru*]-Ø]_{NOMINALISED_TOPIC}]-Ø]_{PREMISE}
 → [[[[...]_{COMMENT} + =*zo*/=*namu*]_{FOCUS} + [[[... *Vstem-ru*]-Ø]_{NOMINALISED_TOPIC}]_{BACKGROUND}]-Ø]_{PREMISE}

Thus, the information-structural complexity is different. In the =*koso* type of *kakari-musubi*, the derivation proceeds in the order: 'clause construction → premise-setting → internal focus–background structuring'. In the =*zo*/=*namu* type, on the other hand, the derivation proceeds in the more complex order: 'topic–comment structure → clause construction → topic-phrase nominalisation → premise-setting → internal focus–background structuring through the addition of =*zo*/=*namu* to the comment phrase'. Even though the clause whose topic-expressing constituent has been nominalised is set as a premise by -Ø [+PREM] before the focalising operation, the intervening -Ø [+NOM] immediately following -*ru* prevents -Ø [+PREM] from standing in the local morphological relation with -*ru* that is

required to trigger the allomorphy $-ru \rightarrow -re$. Therefore, $-ru$ is not realised as $-re$ ⁵.

Accordingly, $=koso$ occurs with $-re$, whereas $=zo$ and $=namu$ occur with $-ru$, simply because the two types of *kakari-musubi* have different morphosyntactic and information-structural configurations. The particles themselves do not condition the predicate forms. Rather, each focus particle occurs in a structure that, independently of the particle, determines whether $-ru$ is realised as $-re$ or remains $-ru$. This reverses the traditional explanation of *kakari-musubi*, according to which the preceding focus particle would determine the form of the predicate. In the present analysis, the morphosyntactic–semantic and information-structural configuration is constructed FIRST, and only THEN determines both the predicate form and which focus particle may occur. The particle $=zo$ or $=namu$ is used to focalise the comment phrase supplied for a topic phrase that contains the predicate and is therefore nominalised by $-\emptyset$ [+NOM], whereas $=koso$ is used to focalise a constituent not containing the finite predicate within a clause that contains no such nominalised topic phrase and is marked as [+PREM] by $-\emptyset$. The crucial morphological difference is that, in the $=koso$ construction, $-\emptyset$ [+PREM] stands in the local relation with $-ru$ that triggers its realisation as $-re$, whereas, in the $=zo/=namu$ construction, the intervening $-\emptyset$ [+NOM] prevents $-\emptyset$ [+PREM] from entering into this local relation with $-ru$. What has traditionally been regarded as direct concord between a particle and a predicate form is therefore a surface correspondence resulting from their occurrence within the same structure, constructed independently of the so-called concord.

5. Isomorphism between verbal and verbal-adjectival realis forms (*izenkei*)

As stated in Section 2, Makino (2025) argues that inflected forms of verbal adjectives and true verbs exhibit the **basic structure 'radical + thematic suffix + ending'** in both EMJ and CJ. Building on that study, (39) schematises the morphosyntactic structures of selected forms of the VA *takasi* — namely, the conjunctive form *taka-ku*, the conclusive form *taka-si- $\emptyset$ ##*, the adnominal form *taka-ki- $\emptyset$ #NP* and the realis forms *taka-ke-re- $\emptyset$* , *taka-ke-re-ba* and *taka-ke-re-do(mo)* — together with their underlying forms and the corresponding forms of the V *tabu*, where Rad., Them.suff. and Infl.suff. stand for *radical*, *thematic suffix* and *inflexional suffix*, respectively:

⁵ The semantic difference between EMJ *haru=koso ki-nu-re* 'Precisely / nothing other than spring has come' and *haru=zo ki-nu-ru* 'It is precisely spring that has come' may be paraphrased in CJ as *masa-ni haru=ga yat-te ki-ta* and *masa-ni haru(=da), yat-te ki-ta=no=wa*, respectively. Alternatively, the difference may be illustrated by means of a deliberately exaggerated paraphrase in highly colloquial CJ as the opposition between *Haru=ga yat-te ki-ta=yo! Masa-ni haru=ga! Haru=ga! Haru=ga!* 'Spring has come! Precisely spring! Spring! Spring!' and *Yat-te ki-ta=yo! Haru=ga! Haru=ga! Haru=ga!* 'It has come! It is spring! Spring! Spring!'. In the former case, the clause 'spring has come' is already constructed and set as a premise, and the constituent *haru* 'spring' is then focalised by $=koso$ within that premise clause. In the latter case, by contrast, *ki-nu-ru* '(it) has come', nominalised by $-\emptyset$ [+NOM], constitutes the topic, and *haru=zo* '(it is) spring' supplies the focalised comment on that nominalised topic.

(39)

Underlying form				
Stem		Ending		
Rad.	Them.suff.	Infl.suff.	Infl.suff.	
<i>taka-</i>	<i>-ku</i>			
<i>tab-</i>	<i>-e</i>			
<i>taka-</i>	<i>-ku</i>	<i>-ru</i>		##
<i>tab-</i>	<i>-e</i>	<i>-ru</i>		##
<i>taka-</i>	<i>-ku</i>	<i>-ru</i>		#NP
<i>tab-</i>	<i>-e</i>	<i>-ru</i>		#NP
<i>taka-</i>	<i>-ku</i>	<i>-ru</i>	<i>-Ø</i>	
<i>tab-</i>	<i>-e</i>	<i>-ru</i>	<i>-Ø</i>	
<i>taka-</i>	<i>-ku</i>	<i>-ru</i>	<i>-ba</i>	
<i>tab-</i>	<i>-e</i>	<i>-ru</i>	<i>-ba</i>	
<i>taka-</i>	<i>-ku</i>	<i>-ru</i>	<i>-do(mo)</i>	
<i>tab-</i>	<i>-e</i>	<i>-ru</i>	<i>-do(mo)</i>	

Surface form				
Stem		Ending		
Rad.	Them.suff.	Infl.suff.	Infl.suff.	
<i>taka-</i>	<i>-ku</i>			
<i>tab-</i>	<i>-e</i>			
<i>taka-</i>	<i>-si</i>	<i>-Ø</i>		##
<i>tab-</i>	<i>-u</i>	<i>-Ø</i>		##
<i>taka-</i>	<i>-ki</i>	<i>-Ø</i>		#NP
<i>tab-</i>	<i>-u</i>	<i>-ru</i>		#NP
<i>taka-</i>	<i>-ke</i>	<i>-re</i>	<i>-Ø</i>	
<i>tab-</i>	<i>-u</i>	<i>-re</i>	<i>-Ø</i>	
<i>taka-</i>	<i>-ke</i>	<i>-re</i>	<i>-ba</i>	
<i>tab-</i>	<i>-u</i>	<i>-re</i>	<i>-ba</i>	
<i>taka-</i>	<i>-ke</i>	<i>-re</i>	<i>-do(mo)</i>	
<i>tab-</i>	<i>-u</i>	<i>-re</i>	<i>-do(mo)</i>	

The inflected forms of *takasi* and *tabu* in (39) exhibit a high degree of structural similarity. Just as the VA thematic suffix *-ku* shows the allomorphy *-ku* → *-si*, *-ki*, *-ke*, depending on the syntactic environment, the V thematic suffix *-e* also exhibits similar allomorphy, namely *-e* → *-u*. The conjunctive forms of both the VA *takasi* and the V *tabu*, *taka-ku* and *tab-e*, lack any inflexional suffix — a defining feature of the predicative stem. The suffix *-ru*, specified as [+ACT], is realised as phonologically null *-Ø* in the VA conclusive form *taka-si-Ø##*, as well as in the corresponding V form *tab-u-Ø##*. Crucially for the present paper, **the suffix *-ru* is consistently realised as *-re* in premise-setting environments** — namely, **before the suffixes *-Ø* [+PREM], *-ba* [+COND] and *-do(mo)* [+CONC]** — not only in V realis forms but **also in VA realis forms**. Furthermore, the VA realis forms such as *taka-ke-re*, *taka-ke-re-ba* and *taka-ke-re-do(mo)* are the only VA inflected forms in which the suffix *-ru* is realised overtly. The existence of such forms allows us to posit the underlying structure common to the conclusive, adnominal and realis forms as *taka-ku-ru-*. The only instance in which the parallelism breaks down is in the adnominal forms *taka-ki-Ø#NP* and *tab-u-ru#NP*, but there seems to be a reason for this. In the VA *takasi*, the opposition between the conclusive and adnominal forms was maintained by the alternation of the thematic-suffix allomorphs *-si* ≈ *-ki*, as in *taka-si-Ø##* and *taka-ki-Ø#NP*, whereas in the V *tabu*, the thematic-suffix allomorph was the same in both forms, namely *-u*, and the opposition was instead maintained by the alternation of the modal-suffix allomorphs *-Ø* ≈ *-ru*, as in *tab-u-Ø##* and *tab-u-ru#NP*.

6. Feature analysis of the realis form *-sika* and the irrealis form *-se* of the past suffix *-ki*

The suffix *-ru* does not indicate whether the eventuality belongs to the past, nor whether it is affirmative or negative. It presents the eventuality represented by the clause as belonging to the actual world and is specified as [+ACT, 0PAST], rather than [+ACT, -PAST]. Where pastness is not independently established, -

ru and its allomorphs *-Ø* and *-re*, as in *Vstem-ru##*, *Vstem-Ø##*, *Vstem-ru#NP* and *Vstem-re-Ø/-ba/-do(mo)*, receive a non-past interpretation through their paradigmatic opposition to *-ki* and its allomorphs *-si* and *-sika*, as in *Vstem-ki##*, *Vstem-si#NP* and *Vstem-sika-Ø/-ba/-do(mo)*, which constitute portmanteau exponents of [+ACT, +PAST]. Where [+PAST] is independently supplied by another form or recoverable from the context, however, tense-neutral *-ru* can also occur with a past eventuality. The exceptional form *-se* constitutes a portmanteau exponent of [−ACT, +PAST] in *Vstem-se-ba* and shares [−ACT] with the suffix *-Ø*, specified as [−ACT, 0PAST], in *Vstem-Ø-ba*. The feature [+PAST] in *-se*, owing to the co-occurring [−ACT], cannot anchor a past eventuality in the actual world. Consequently, [+PAST] is reinterpreted as [+IRREAL] through the other-world value in the binary world-opposition schema ACTUAL WORLD vs. OTHER WORLD, giving rise to a counterfactual reading. Therefore, the semantic oppositions between *mi-re-ba*, *mi-sika-ba*, *mi-Ø-ba* and *mi-se-ba* from the verb *miru* 'look at' can be analysed as demonstrated in (40):

(40)

	[ACT]	[PAST]	[COND]	Interpretation
<i>mi-re-ba</i>	+	0	+	actual-world cause, occasion or general condition: 'since/when/whenever one sees'
<i>mi-sika-ba</i>	+	+	+	actual-world past cause or occasion: 'since/when one saw'
<i>mi-Ø-ba</i>	−	0	+	hypothetical condition: 'if one sees'
<i>mi-se-ba</i>	−	+	+	counterfactual condition: 'if one saw/were to see; if one had seen'

Note that the suffix *-ki* is said to be often used to 'express the memory of one's own experience' (Ōno et al. 1990: 1481, translated from Japanese). This may be because direct experience constitutes the most immediate epistemic ground for attributing a past eventuality to the actual world, and therefore typically licenses *-ki*, specified as [+ACT, +PAST].

7. Reanalysis of the suffix–suffix sequence *-re-ba* into the single conditional suffix *-reba*

In EMJ, the morphosyntactic–semantic opposition in (41) was still maintained:

- (41) UF *yom-Ø-Ø*_[−ACT]-*ba* → SF *yom-a-Ø-ba* 'if one reads'
 UF *yom-Ø-ru*_[+ACT]-*ba* → IF *yom-Ø-re-ba* → SF *yom-Ø-e-ba* 'since/when one reads'

However, in the course of the development towards CJ, in accordance with the tendency for the V thematic suffix *-Ø* to be realised, in principle, as phonologically null, forms containing its allomorph *-a*, such as *yom-a-Ø-ba*, fell out of use, except in fixed expressions such as *suki ar-a-Ø-ba* 'given half a chance' and literary expressions such as *sin-a-Ø-ba moro=tomo* 'if we die, let us die together'. As a result,

the productive morphosyntactic–semantic opposition between the realis conditional form in *-re*_[+ACT]-*ba* and the irrealis conditional form in *-Ø*_[−ACT]-*ba* was lost. Once *-re* no longer had its productive oppositional counterpart *-Ø*, specified as [−ACT], within the conditional paradigm, it also lost its semantic value [+ACT]. This is because linguistic value arises from opposition, as Saussure (1916: 187, 196) puts it: "Dans l'intérieur d'une même langue, tous les mots qui expriment des idées voisines se limitent réciproquement ... n'ont de valeur propre que par leur opposition"; "deux signes comportant chacun un signifié et un signifiant ... sont seulement distincts. Entre eux il n'y a qu'opposition. Tout le mécanisme du langage ... repose sur des oppositions de ce genre et sur les différences phoniques et conceptuelles qu'elles impliquent".

Having lost its semantic value, *-re* was no longer recognised as an independent morpheme and, owing to the resulting opacity of the morpheme boundary, merged with the following *-ba* to form the single morpheme *-reba*. This is the CJ conditional suffix *-reba*. Since the reanalysed *-reba* has lost the [+ACT] component originally possessed by *-re* as an allomorph of *-ru*, its meaning has changed from [+ACT, +COND] to [+COND]. Consequently, although it may still receive an actual-world causal, occasional or general-condition interpretation where this is contextually supported, it no longer marks such an interpretation through a [+ACT] component and has become available for hypothetical conditions. The derivation of the SF *yomeba* in EMJ and CJ can be schematised as in (42):

- (42) EMJ: UF *yom-Ø-ru-ba* → IF *yom-Ø-re-ba* → SF *yom-Ø-e-ba* 'since/when one reads'
 CJ: UF *yom-Ø-reba* → SF *yom-Ø-eba* 'if one reads'

This diachronic scenario awaits fuller philological verification, but it appears to provide a coherent account of the reanalysis of *-re-ba* into CJ *-reba*.

8. Derivation of verbal realis and irrealis forms in EMJ

As noted in Sections 2 and 3, the V thematic suffixes *-i* and *-e* of multigrade vowel-stem verbs and the phonologically null V thematic suffix *-Ø* of consonant-stem verbs exhibit the allomorphies in (43) (= (14), (15), (22), (25) and (26)):

- (43) V thematic suffixes *-i* and *-e* → *-u* / *-ru* in multigrade vowel-stem verbs

V thematic suffix *-Ø* → *-u* / *-ru* in trigrade consonant-stem verbs

V thematic suffix *-Ø* → *-a* / *-Ø* [−ACT], *-zu*, *-mu*, *-zi*, *-masi*, *-mahosi* }

V thematic suffix *-i* → *-e* / *-Ø* [−ACT], *-zu*, *-mu*, *-zi*, *-masi*, *-mahosi*-, *-yo* } in the *s*-irregular verb

V thematic suffix *-i* → *-o* / $_ \{ -\emptyset [-\text{ACT}], -zu, -mu, -zi, -masi, -mahosi-, -yo \}$ in the *k*-irregular verb

As discussed in Sections 2 and 4, moreover, the suffix *-ru*, specified as [+ACT], shows the allomorphies in (44) (= (16) and (36)):

- (44) *-ru* → $\emptyset$ / $_\#$ in trigrade consonant-stem verbs, multigrade vowel-stem verbs and *r*-irregular verbs
-ru → *-re* / $_ \{ -\emptyset [+PREM], -ba, -do(mo) \}$

Traditionally, the so-called *izenkei* — realis forms ending underlyingly in *-ru- $\emptyset$* [+PREM], *-ru-ba* and *-ru-do(mo)* — have been distinguished from the so-called *syūsikei* — conclusive forms ending underlyingly in *-ru###* — and the so-called *rentaikei* — adnominal forms ending underlyingly in *-ru#NP*. However, insofar as they all share the underlying suffix *-ru*, which marks a clause as [+ACT] and thereby presents the eventuality expressed by that clause as belonging to the actual world, they are equally realis forms. On the other hand, the so-called *mizenkei* forms are properly analysed as irrealis forms, in that they contain, instead of *-ru*, the irrealis set $\{ -\emptyset [-\text{ACT}], -zu, -mu, -zi, -masi, -mahosi- \}$, which stands in modal opposition to the realis set $\{ -ru [+ACT] \}$. Building on the classical yet fundamental Saussurean thesis that **linguistic value arises from opposition** (cf. Section 7), it is more plausible to assume that **EMJ *-ru*, unlike CJ *-ru*, had a strong [+ACT] value within its opposition to the rich series of irrealis suffixes, including the particularly frequent *-zu* and *-mu***. This is because the semantic content of a morpheme is determined precisely within the system of terms with which it stands in opposition. This point has not, however, been sufficiently taken into account in previous treatments of the value of EMJ *-ru*.

From the perspective of the present analysis, the traditional conjugational table of the so-called Hashimoto grammar conflates two distinct dimensions. Forms traditionally classified as conclusive, adnominal and realis share the underlying suffix *-ru* [+ACT], although it surfaces as $\emptyset$, *-ru* or *-re* depending on the local environment. On the other hand, the forms traditionally classified as irrealis, conjunctive and imperative lack this suffix and instead involve either the $[-\text{ACT}]$ series $\{ -\emptyset [-\text{ACT}], -zu, -mu, -zi, -masi, -mahosi-, -yo \}$ or non-modal environments. The fundamental difficulty of the traditional analysis thus lies in its failure to recognise *-ru* as an independent morpheme specified as [+ACT]. Once *-ru* is recognised as an independent [+ACT] morpheme, the traditional opposition among *syūsikei*, *rentaikei* and *izenkei* is reinterpreted as a surface classification of different allomorphs of the same underlying realis structure.

The realis forms reanalysed above, such as *kik-u###* 'one hears', *kik-u#NP* 'NP that one hears', ...*koso* ... *kik-e* 'one hears ... exactly ...', *kik-e-ba* 'since/when one hears' and *kik-e-do(mo)* 'though one hears', and the irrealis forms, such as *kik-a-ba* 'if one hears', *kik-a-zu* 'one does not hear', *kik-a-mu* 'one will hear', *kik-a-zi* 'one will not hear' and *kik-a-masi* 'one would hear', in which, at first sight, *-e* and *-a* appear to stand

in formal and semantic opposition, are derived straightforwardly from the corresponding UFs through the allomorphies recapitulated in (43) and (44) at the beginning of this section, together with deletion of the suffix-initial *r*- after a stem-final consonant. This is illustrated in (45) with *kiku* and other verb stems:

(45) UF		IF		SF
<i>kik-</i> $\emptyset$		$\rightarrow$ <i>kik-</i> $\emptyset$		$\rightarrow$ <i>kik</i>
<i>kik-</i> $\emptyset$ <i>-ru</i> ##		$\rightarrow$ <i>kik-</i> $\emptyset$ <i>-ru</i> ##		$\rightarrow$ <i>kik-u##</i>
<i>kik-</i> $\emptyset$ <i>-ru</i> #NP		$\rightarrow$ <i>kik-</i> $\emptyset$ <i>-ru</i> #NP		$\rightarrow$ <i>kik-u#NP</i>
<i>kik-</i> $\emptyset$ <i>-ru</i> $\emptyset$		$\rightarrow$ <i>kik-</i> $\emptyset$ <i>-re</i> $\emptyset$		$\rightarrow$ <i>kik-e</i>
<i>kik-</i> $\emptyset$ <i>-ru</i> <i>-ba</i>		$\rightarrow$ <i>kik-</i> $\emptyset$ <i>-re</i> <i>-ba</i>		$\rightarrow$ <i>kik-e-ba</i>
<i>kik-</i> $\emptyset$ <i>-ru</i> <i>-do(mo)</i>		$\rightarrow$ <i>kik-</i> $\emptyset$ <i>-re</i> <i>-do(mo)</i>		$\rightarrow$ <i>kik-e-do(mo)</i>
<i>kik-</i> $\emptyset$ $\emptyset$ <i>-ba</i>		$\rightarrow$ <i>kik-</i> <i>-a</i> $\emptyset$ <i>-ba</i>		$\rightarrow$ <i>kik-a-ba</i>
<i>kik-</i> $\emptyset$ <i>-zu</i>		$\rightarrow$ <i>kik-</i> <i>-a</i> <i>-zu</i>		$\rightarrow$ <i>kik-a-zu</i>
<i>kik-</i> $\emptyset$ <i>-mu</i>		$\rightarrow$ <i>kik-</i> <i>-a</i> <i>-mu</i>		$\rightarrow$ <i>kik-a-mu</i>
<i>kik-</i> $\emptyset$ <i>-zi</i>		$\rightarrow$ <i>kik-</i> <i>-a</i> <i>-zi</i>		$\rightarrow$ <i>kik-a-zi</i>
<i>kik-</i> $\emptyset$ <i>-masi</i>		$\rightarrow$ <i>kik-</i> <i>-a</i> <i>-masi</i>		$\rightarrow$ <i>kik-a-masi</i>
<i>kik-</i> $\emptyset$ <i>-mahosi-</i>		$\rightarrow$ <i>kik-</i> <i>-a</i> <i>-mahosi-</i>		$\rightarrow$ <i>kik-a-mahosi-</i>
<i>kik-</i> $\emptyset$ <i>-X</i>		$\rightarrow$ <i>kik-</i> $\emptyset$ <i>-X</i>		$\rightarrow$ <i>kik-e</i>
<i>ar-</i> $\emptyset$		$\rightarrow$ <i>ar-</i> $\emptyset$		$\rightarrow$ <i>ari</i>
<i>ar-</i> $\emptyset$ <i>-ru</i> ##		$\rightarrow$ <i>ar-</i> $\emptyset$ $\emptyset$ ##		$\rightarrow$ <i>ari##</i>
<i>ar-</i> $\emptyset$ <i>-ru</i> #NP		$\rightarrow$ <i>ar-</i> $\emptyset$ <i>-ru</i> #NP		$\rightarrow$ <i>ar-u#NP</i>
<i>ar-</i> $\emptyset$ <i>-ru</i> $\emptyset$		$\rightarrow$ <i>ar-</i> $\emptyset$ <i>-re</i> $\emptyset$		$\rightarrow$ <i>ar-e</i>
<i>ar-</i> $\emptyset$ <i>-ru</i> <i>-ba</i>		$\rightarrow$ <i>ar-</i> $\emptyset$ <i>-re</i> <i>-ba</i>		$\rightarrow$ <i>ar-e-ba</i>
<i>ar-</i> $\emptyset$ <i>-ru</i> <i>-do(mo)</i>		$\rightarrow$ <i>ar-</i> $\emptyset$ <i>-re</i> <i>-do(mo)</i>		$\rightarrow$ <i>ar-e-do(mo)</i>
<i>ar-</i> $\emptyset$ $\emptyset$ <i>-ba</i>		$\rightarrow$ <i>ar-</i> <i>-a</i> $\emptyset$ <i>-ba</i>		$\rightarrow$ <i>ar-a-ba</i>
<i>ar-</i> $\emptyset$ <i>-zu</i>		$\rightarrow$ <i>ar-</i> <i>-a</i> <i>-zu</i>		$\rightarrow$ <i>ar-a-zu</i>
<i>ar-</i> $\emptyset$ <i>-mu</i>		$\rightarrow$ <i>ar-</i> <i>-a</i> <i>-mu</i>		$\rightarrow$ <i>ar-a-mu</i>
<i>ar-</i> $\emptyset$ <i>-zi</i>		$\rightarrow$ <i>ar-</i> <i>-a</i> <i>-zi</i>		$\rightarrow$ <i>ar-a-zi</i>
<i>ar-</i> $\emptyset$ <i>-masi</i>		$\rightarrow$ <i>ar-</i> <i>-a</i> <i>-masi</i>		$\rightarrow$ <i>ar-a-masi</i>
<i>ar-</i> $\emptyset$ <i>-mahosi-</i>		$\rightarrow$ <i>ar-</i> <i>-a</i> <i>-mahosi-</i>		$\rightarrow$ <i>ar-a-mahosi-</i>
<i>ar-</i> $\emptyset$ <i>-X</i>		$\rightarrow$ <i>ar-</i> $\emptyset$ <i>-X</i>		$\rightarrow$ <i>ar-e</i>
<i>sin-</i> $\emptyset$		$\rightarrow$ <i>sin-</i> $\emptyset$		$\rightarrow$ <i>sini</i>
<i>sin-</i> $\emptyset$ <i>-ru</i> ##		$\rightarrow$ <i>sin-</i> <i>-u</i> $\emptyset$ ##		$\rightarrow$ <i>sin-u##</i>
<i>sin-</i> $\emptyset$ <i>-ru</i> #NP		$\rightarrow$ <i>sin-</i> <i>-u</i> <i>-ru</i> #NP		$\rightarrow$ <i>sin-u-ru#NP</i>
<i>sin-</i> $\emptyset$ <i>-ru</i> $\emptyset$		$\rightarrow$ <i>sin-</i> <i>-u</i> <i>-re</i> $\emptyset$		$\rightarrow$ <i>sin-u-re</i>
<i>sin-</i> $\emptyset$ <i>-ru</i> <i>-ba</i>		$\rightarrow$ <i>sin-</i> <i>-u</i> <i>-re</i> <i>-ba</i>		$\rightarrow$ <i>sin-u-re-ba</i>
<i>sin-</i> $\emptyset$ <i>-ru</i> <i>-do(mo)</i>		$\rightarrow$ <i>sin-</i> <i>-u</i> <i>-re</i> <i>-do(mo)</i>		$\rightarrow$ <i>sin-u-re-do(mo)</i>
<i>sin-</i> $\emptyset$ $\emptyset$ <i>-ba</i>		$\rightarrow$ <i>sin-</i> <i>-a</i> $\emptyset$ <i>-ba</i>		$\rightarrow$ <i>sin-a-ba</i>
<i>sin-</i> $\emptyset$ <i>-zu</i>		$\rightarrow$ <i>sin-</i> <i>-a</i> <i>-zu</i>		$\rightarrow$ <i>sin-a-zu</i>
<i>sin-</i> $\emptyset$ <i>-mu</i>		$\rightarrow$ <i>sin-</i> <i>-a</i> <i>-mu</i>		$\rightarrow$ <i>sin-a-mu</i>
<i>sin-</i> $\emptyset$ <i>-zi</i>		$\rightarrow$ <i>sin-</i> <i>-a</i> <i>-zi</i>		$\rightarrow$ <i>sin-a-zi</i>
<i>sin-</i> $\emptyset$ <i>-masi</i>		$\rightarrow$ <i>sin-</i> <i>-a</i> <i>-masi</i>		$\rightarrow$ <i>sin-a-masi</i>
<i>sin-</i> $\emptyset$ <i>-mahosi-</i>		$\rightarrow$ <i>sin-</i> <i>-a</i> <i>-mahosi-</i>		$\rightarrow$ <i>sin-a-mahosi-</i>
<i>sin-</i> $\emptyset$ <i>-X</i>		$\rightarrow$ <i>sin-</i> $\emptyset$ <i>-X</i>		$\rightarrow$ <i>sin-e</i>
<i>S-</i> <i>-i</i>		$\rightarrow$ <i>S-</i> <i>-i</i>		$\rightarrow$ <i>S-i</i>
<i>S-</i> <i>-i</i> <i>-ru</i> ##		$\rightarrow$ <i>S-</i> <i>-u</i> $\emptyset$ ##		$\rightarrow$ <i>S-u##</i>

S- -i -ru	#NP	→	S- -u -ru	#NP	→	S-u-ru#NP
S- -i -ru	-Ø	→	S- -u -re	-Ø	→	S-u-re
S- -i -ru	-ba	→	S- -u -re	-ba	→	S-u-re-ba
S- -i -ru	-do(mo)	→	S- -u -re	-do(mo)	→	S-u-re-do(mo)
S- -i -Ø	-ba	→	S- -e -Ø	-ba	→	S-e-ba
S- -i -zu		→	S- -e -zu		→	S-e-zu
S- -i -mu		→	S- -e -mu		→	S-e-mu
S- -i -zi		→	S- -e -zi		→	S-e-zi
S- -i -masi		→	S- -e -masi		→	S-e-masi
S- -i -mahosi-		→	S- -e -mahosi-		→	S-e-mahosi-
S- -i -yo		→	S- -e -yo		→	S-e-yo
K- -i		→	K- -i		→	K-i
K- -i -ru	##	→	K- -u -Ø	##	→	K-u##
K- -i -ru	#NP	→	K- -u -ru	#NP	→	K-u-ru#NP
K- -i -ru	-Ø	→	K- -u -re	-Ø	→	K-u-re
K- -i -ru	-ba	→	K- -u -re	-ba	→	K-u-re-ba
K- -i -ru	-do(mo)	→	K- -u -re	-do(mo)	→	K-u-re-do(mo)
K- -i -Ø	-ba	→	K- -o -Ø	-ba	→	K-o-ba
K- -i -zu		→	K- -o -zu		→	K-o-zu
K- -i -mu		→	K- -o -mu		→	K-o-mu
K- -i -zi		→	K- -o -zi		→	K-o-zi
K- -i -masi		→	K- -o -masi		→	K-o-masi
K- -i -mahosi-		→	K- -o -mahosi-		→	K-o-mahosi-
K- -i -yo		→	K- -o -yo		→	K-o-yo
nob- -i		→	nob- -i		→	nob-i
nob- -i -ru	##	→	nob- -u -Ø	##	→	nob-u##
nob- -i -ru	#NP	→	nob- -u -ru	#NP	→	nob-u-ru#NP
nob- -i -ru	-Ø	→	nob- -u -re	-Ø	→	nob-u-re
nob- -i -ru	-ba	→	nob- -u -re	-ba	→	nob-u-re-ba
nob- -i -ru	-do(mo)	→	nob- -u -re	-do(mo)	→	nob-u-re-do(mo)
nob- -i -Ø	-ba	→	nob- -i -Ø	-ba	→	nob-i-ba
nob- -i -zu		→	nob- -i -zu		→	nob-i-zu
nob- -i -mu		→	nob- -i -mu		→	nob-i-mu
nob- -i -zi		→	nob- -i -zi		→	nob-i-zi
nob- -i -masi		→	nob- -i -masi		→	nob-i-masi
nob- -i -mahosi-		→	nob- -i -mahosi-		→	nob-i-mahosi-
nob- -i -yo		→	nob- -i -yo		→	nob-i-yo
nob- -e		→	nob- -e		→	nob-e
nob- -e -ru	##	→	nob- -u -Ø	##	→	nob-u##
nob- -e -ru	#NP	→	nob- -u -ru	#NP	→	nob-u-ru#NP
nob- -e -ru	-Ø	→	nob- -u -re	-Ø	→	nob-u-re
nob- -e -ru	-ba	→	nob- -u -re	-ba	→	nob-u-re-ba
nob- -e -ru	-do(mo)	→	nob- -u -re	-do(mo)	→	nob-u-re-do(mo)
nob- -e -Ø	-ba	→	nob- -e -Ø	-ba	→	nob-e-ba
nob- -e -zu		→	nob- -e -zu		→	nob-e-zu
nob- -e -mu		→	nob- -e -mu		→	nob-e-mu
nob- -e -zi		→	nob- -e -zi		→	nob-e-zi
nob- -e -masi		→	nob- -e -masi		→	nob-e-masi

<i>nob-</i>	<i>-e</i>	<i>-mahosi-</i>		→	<i>nob-</i>	<i>-e</i>	<i>-mahosi-</i>		→	<i>nob-e-mahosi-</i>
<i>nob-</i>	<i>-e</i>	<i>-yo</i>		→	<i>nob-</i>	<i>-e</i>	<i>-yo</i>		→	<i>nob-e-yo</i>
<i>m-</i>	<i>-i</i>			→	<i>m-</i>	<i>-i</i>			→	<i>m-i</i>
<i>m-</i>	<i>-i</i>	<i>-ru</i>	##	→	<i>m-</i>	<i>-i</i>	<i>-ru</i>	##	→	<i>m-i-ru##</i>
<i>m-</i>	<i>-i</i>	<i>-ru</i>	##	→	<i>m-</i>	<i>-i</i>	<i>-ru</i>	##	→	<i>m-i-ru##</i>
<i>m-</i>	<i>-i</i>	<i>-ru</i>	#NP	→	<i>m-</i>	<i>-i</i>	<i>-ru</i>	#NP	→	<i>m-i-ru#NP</i>
<i>m-</i>	<i>-i</i>	-ru	-∅	→	<i>m-</i>	<i>-i</i>	-re	-∅	→	<i>m-i-re</i>
<i>m-</i>	<i>-i</i>	-ru	-ba	→	<i>m-</i>	<i>-i</i>	-re	-ba	→	<i>m-i-re-ba</i>
<i>m-</i>	<i>-i</i>	-ru	-do(mo)	→	<i>m-</i>	<i>-i</i>	-re	-do(mo)	→	<i>m-i-re-do(mo)</i>
<i>m-</i>	<i>-i</i>	-∅	-ba	→	<i>m-</i>	<i>-i</i>	-∅	-ba	→	<i>m-i-ba</i>
<i>m-</i>	<i>-i</i>	<i>-zu</i>		→	<i>m-</i>	<i>-i</i>	<i>-zu</i>		→	<i>m-i-zu</i>
<i>m-</i>	<i>-i</i>	<i>-mu</i>		→	<i>m-</i>	<i>-i</i>	<i>-mu</i>		→	<i>m-i-mu</i>
<i>m-</i>	<i>-i</i>	<i>-zi</i>		→	<i>m-</i>	<i>-i</i>	<i>-zi</i>		→	<i>m-i-zi</i>
<i>m-</i>	<i>-i</i>	<i>-masi</i>		→	<i>m-</i>	<i>-i</i>	<i>-masi</i>		→	<i>m-i-masi</i>
<i>m-</i>	<i>-i</i>	<i>-mahosi-</i>		→	<i>m-</i>	<i>-i</i>	<i>-mahosi-</i>		→	<i>m-i-mahosi-</i>
<i>m-</i>	<i>-i</i>	<i>-yo</i>		→	<i>m-</i>	<i>-i</i>	<i>-yo</i>		→	<i>m-i-yo</i>
<i>k-</i>	<i>-e</i>			→	<i>k-</i>	<i>-e</i>			→	<i>k-e</i>
<i>k-</i>	<i>-e</i>	<i>-ru</i>	##	→	<i>k-</i>	<i>-e</i>	<i>-ru</i>	##	→	<i>k-e-ru##</i>
<i>k-</i>	<i>-e</i>	<i>-ru</i>	#NP	→	<i>k-</i>	<i>-e</i>	<i>-ru</i>	#NP	→	<i>k-e-ru#NP</i>
<i>k-</i>	<i>-e</i>	-ru	-∅	→	<i>k-</i>	<i>-e</i>	-re	-∅	→	<i>k-e-re</i>
<i>k-</i>	<i>-e</i>	-ru	-ba	→	<i>k-</i>	<i>-e</i>	-re	-ba	→	<i>k-e-re-ba</i>
<i>k-</i>	<i>-e</i>	-ru	-do(mo)	→	<i>k-</i>	<i>-e</i>	-re	-do(mo)	→	<i>k-e-re-do(mo)</i>
<i>k-</i>	<i>-e</i>	-∅	-ba	→	<i>k-</i>	<i>-e</i>	-∅	-ba	→	<i>k-e-ba</i>
<i>k-</i>	<i>-e</i>	<i>-zu</i>		→	<i>k-</i>	<i>-e</i>	<i>-zu</i>		→	<i>k-e-zu</i>
<i>k-</i>	<i>-e</i>	<i>-mu</i>		→	<i>k-</i>	<i>-e</i>	<i>-mu</i>		→	<i>k-e-mu</i>
<i>k-</i>	<i>-e</i>	<i>-zi</i>		→	<i>k-</i>	<i>-e</i>	<i>-zi</i>		→	<i>k-e-zi</i>
<i>k-</i>	<i>-e</i>	<i>-masi</i>		→	<i>k-</i>	<i>-e</i>	<i>-masi</i>		→	<i>k-e-masi</i>
<i>k-</i>	<i>-e</i>	<i>-mahosi-</i>		→	<i>k-</i>	<i>-e</i>	<i>-mahosi-</i>		→	<i>k-e-mahosi-</i>
<i>k-</i>	<i>-e</i>	<i>-yo</i>		→	<i>k-</i>	<i>-e</i>	<i>-yo</i>		→	<i>k-e-yo</i>

Thus, the two phonologically null suffixes -Ø [-ACT] and -Ø [+PREM] are independently supported by the same diagnostic: each forms a semantically coherent natural class with other overt suffixes — respectively, -Ø [-ACT] with *-zu*, *-mu*, *-zi*, *-masi*, *-mahosi-* and *-yo*, which, in opposition to *-ru* [+ACT], do not anchor the eventuality expressed by the verbal stem together with its argument(s) to the actual world, and -Ø [+PREM] with *-ba* and *-do(mo)*, which incorporate a clause as a premise into another semantic structure — and each locally conditions the allomorphy of the immediately preceding morpheme: the V thematic suffixes -Ø → *-a*, *-i* → *-e* and *-i* → *-o* by -Ø [-ACT], and the suffix *-ru* → *-re* by -Ø [+PREM]. It follows, therefore, that the domains traditionally treated separately — the *=koso* type of *kakari-musubi*, conditional expressions and the irrealis–realis opposition — can be reduced to three semantic natural classes, the realis {-*ru* [+ACT]}, the irrealis {-Ø [-ACT], *-zu*, *-mu*, *-zi*, *-masi*, *-mahosi-*, *-yo*} and the premise-setting {-Ø [+PREM], *-ba*, *-do(mo)*}, and seven local allomorphic alternations in (43) and (44).

Under this analysis, the traditional classes labelled *sahen*, *kahen* and *nahen* are not irregular in their

morphophonological behaviour. Rather, *sahen* and *kahen* are trigrade vowel-stem verbs, while *nahen* verbs are trigrade consonant-stem verbs. In the former two classes, the V thematic suffix *-i* alternates with *-u* and with *-e* or *-o*, respectively; in *nahen* verbs, the phonologically null V thematic suffix $-\emptyset$ alternates with *-a* and *-u*. (Note that the vowel *i* in forms such as *sini-te* is excluded from this grade alternation, since it is analysed as epenthetic; see footnote 2.) These alternations, $-i \approx -u \approx -e$, $-i \approx -u \approx -o$ and $-\emptyset \approx -u \approx -a$, respectively, are semantically coherent, as illustrated in (46) and (47):

(46)

UF	SF: realis environment	SF: irrealis environment	SF: Elsewhere	Stem class
<i>-i</i>	<i>-i</i>			Monograde vowel-stem
<i>-e</i>	<i>-e</i>			Monograde vowel-stem
<i>-i</i>	<i>-u</i>	<i>-i</i>		Bigrade vowel-stem
<i>-e</i>	<i>-u</i>	<i>-e</i>		Bigrade vowel-stem
<i>-i</i>	<i>-u</i>	<i>-e</i>	<i>-i</i>	Trigrade vowel-stem
<i>-i</i>	<i>-u</i>	<i>-o</i>	<i>-i</i>	Trigrade vowel-stem

(47)

UF	SF: irrealis environment	SF: realis environment	SF: Elsewhere	Stem class
$-\emptyset$	<i>-a</i>	$-\emptyset$		Bigrade consonant-stem
$-\emptyset$	<i>-a</i>	<i>-u</i>	$-\emptyset$	Trigrade consonant-stem

Bigrade verbs merely differentiate stem grades according to the following morphosyntactic–semantic environment where monograde verbs leave them undifferentiated, and trigrade verbs likewise differentiate stem grades where bigrade verbs leave them undifferentiated. Consequently, apart from this difference in grade, their morphophonological behaviour is governed by the same type of local allomorphic alternation as that found in the other verb classes. What has traditionally been treated as irregularity is reanalysed here as **semantically coherent grade differentiation conditioned by the following modal environment**.

It should further be noted that, just as CJ speakers can predict, when they hear *kik-a-* from the verb *kiku*, that the following element will be a negative form such as *-nai* or *-zu=ni*, EMJ speakers may also have been able, to some extent, to predict the following modal environment from the alternations of the V thematic suffix, although the details differed according to stem class. With regard to the $-i/-e \rightarrow -u$ alternation in bigrade vowel-stem verbs, Hayata (2000: 25), drawing on the reconstructed pre-literary antecedents of *-ru* and *-re*, argues: "The reconstructed forms **rua* (adnominal) and **rua-gi* (conditional) through comparison between the Central dialects and the Eastern dialects of the Nara Period, and through internal reconstruction, etc. appear to account for the alternation, since the alternation occurs only before segments beginning with /ru/ but not before segments beginning with /ra/. The alternation suggests regressive assimilation in verb with stem final oturai vowels." The phonological origin of this alternation does not preclude, however, its subsequent morphosyntactic–semantic reinterpretation: once grammaticalised, the alternation may have functioned as a co-marker formally distinguishing the *-ru*

[+ACT] class. Moreover, the occurrence of the allomorphy $-ru \rightarrow -\emptyset / _##$ in multigrade vowel-stem verbs and trigrade consonant-stem verbs, as in UF $nob-i-ru## \rightarrow$ IF $nob-u-ru## \rightarrow$ SF $nob-u-\emptyset##$, may also be attributed to the preceding allomorphy $-i, -e, -\emptyset \rightarrow -u / _ru$, which makes the underlying $-ru$ recoverable even after its deletion and allows the conclusive and adnominal forms to be formally distinguished at the surface level, as in $nob-u##$ vs. $nob-u-ru##$ NP. The tendency towards a reduction in the number of grades in the development from EMJ to CJ may therefore be correlated with the decline of suffixal modality marking, since the loss of grade distinctions may reflect not merely morphophonological simplification, but also the weakening of a system in which stem-grade alternations helped to signal the modal environment supplied by the following suffix.

What remains genuinely irregular under the present analysis is essentially limited to the conclusive-form ending of the r -irregular verbs and to the formation of imperative forms of consonant-stem verbs, where the expected form $*kik-a-yo$, if the imperative suffix were $-yo$, is not attested for the verb *kiku*, for instance. Consequently, the morphosyntactic–semantic verbal system of EMJ need not be organised around the traditional conjugational classes, but can instead be reorganised in terms of the following oppositions:

- **Consonant-stem vs. vowel-stem verbs**
- **Monograde, bigrade and trigrade patterns**
- **Regularity vs. residual irregularity**

Note finally that the other EMJ modal suffixes $-ramu$, $-rasi$, $-mer$, $-nar$ – $-be$ and $-mazi$ -, as in $m-i-ru-ramu$ 'one will be seeing about now (out of the speaker's sight)', $m-i-ru-rasi$ 'one must see (given the evidence)', $m-i-ru-meri$ 'one seems to see', $m-i-ru-nari$ 'one is said to see / one seems to see (on auditory evidence)', $m-i-ru-be-si$ 'one must/should see' and $m-i-ru-mazi$ 'one must/should not see', occur in a position distinct from that of $-ru$ in the syntactic structure and therefore do not stand in direct paradigmatic opposition to $-ru$, as illustrated in (48):

(48) UF	IF	SF
$m- -i -ru$	$\rightarrow m- -i -ru$	$\rightarrow m-i-ru$
$m- -i -mu$	$\rightarrow m- -i -mu$	$\rightarrow m-i-mu$
$m- -i -ru -ramu$	$\rightarrow m- -i -ru -ramu$	$\rightarrow m-i-ru-ramu$
$m- -i -ru -rasi$	$\rightarrow m- -i -ru -rasi$	$\rightarrow m-i-ru-rasi$
$m- -i -ru -mer -\emptyset -ru$	$\rightarrow m- -i -ru -mer -\emptyset -\emptyset$	$\rightarrow m-i-ru-meri$
$m- -i -ru -nar -\emptyset -ru$	$\rightarrow m- -i -ru -nar -\emptyset -\emptyset$	$\rightarrow m-i-ru-nari$
$m- -i -ru -be -ku -ru$	$\rightarrow m- -i -ru -be -si -\emptyset$	$\rightarrow m-i-ru-be-si$
$m- -i -ru -mazi -ku -ru$	$\rightarrow m- -i -ru -mazi -\emptyset -\emptyset$	$\rightarrow m-i-ru-mazi$

Whereas *-ru* and *-mu* occupy the same syntactic-structural position, namely that of the inflexional suffix immediately following the verbal stem *m-i(-)*, and therefore stand in direct paradigmatic opposition, the suffixes *-ramu*, *-rasi*, *-mer-*, *-nar-*, *-be-* and *-mazi-* occur immediately outside the inflected form already completed by the addition of *-ru* to the verbal stem *m-i(-)*. Taking the auditory evidential suffix *-nar-* as an example, the preceding *-ru* shows the same morphophonological behaviour as *-ru* immediately preceding the main-clause-final boundary *##*, as indicated in (49):

(49) UF		IF		IF		SF
<i>m- -i -ru -nar- -Ø -ru</i>	→	<i>m- -i -ru -nar- -Ø -Ø</i>	→	<i>m-i-ru-nar</i>	→	<i>m-i-ru-nari</i>
<i>k- -e -ru -nar- -Ø -ru</i>	→	<i>k- -e -ru -nar- -Ø -Ø</i>	→	<i>k-e-ru-nar</i>	→	<i>k-e-ru-nari</i>
<i>nob- -i -ru -nar- -Ø -ru</i>	→	<i>nob- -u -Ø -nar- -Ø -Ø</i>	→	<i>nob-u-nar</i>	→	<i>nob-u-nari</i>
<i>nob- -e -ru -nar- -Ø -ru</i>	→	<i>nob- -u -Ø -nar- -Ø -Ø</i>	→	<i>nob-u-nar</i>	→	<i>nob-u-nari</i>
<i>S- -i -ru -nar- -Ø -ru</i>	→	<i>S- -u -Ø -nar- -Ø -Ø</i>	→	<i>S-u-nar</i>	→	<i>S-u-nari</i>
<i>K- -i -ru -nar- -Ø -ru</i>	→	<i>K- -u -Ø -nar- -Ø -Ø</i>	→	<i>K-u-nar</i>	→	<i>K-u-nari</i>
<i>kik- -Ø -ru -nar- -Ø -ru</i>	→	<i>kik- -Ø -ru -nar- -Ø -Ø</i>	→	<i>kik-ru-nar</i>	→	<i>kik-u-nari</i>
<i>sin- -Ø -ru -nar- -Ø -ru</i>	→	<i>sin- -u -Ø -nar- -Ø -Ø</i>	→	<i>sin-u-nar</i>	→	<i>sin-u-nari</i>
<i>ar- -Ø -ru -nar- -Ø -ru</i>	→	<i>ar- -Ø -Ø -nar- -Ø -Ø</i>	→	<i>ar-nar</i>	→	<i>ari-nari</i> ⁶
<i>m- -i -ru ##</i>	→	<i>m- -i -ru ##</i>	→	<i>m-i-ru</i>	→	<i>m-i-ru</i>
<i>k- -e -ru ##</i>	→	<i>k- -e -ru ##</i>	→	<i>k-e-ru</i>	→	<i>k-e-ru</i>
<i>nob- -i -ru ##</i>	→	<i>nob- -u -Ø ##</i>	→	<i>nob-u</i>	→	<i>nob-u</i>
<i>nob- -e -ru ##</i>	→	<i>nob- -u -Ø ##</i>	→	<i>nob-u</i>	→	<i>nob-u</i>
<i>S- -i -ru ##</i>	→	<i>S- -u -Ø ##</i>	→	<i>S-u</i>	→	<i>S-u</i>
<i>K- -i -ru ##</i>	→	<i>K- -u -Ø ##</i>	→	<i>K-u</i>	→	<i>K-u</i>
<i>kik- -Ø -ru ##</i>	→	<i>kik- -Ø -ru ##</i>	→	<i>kik-ru</i>	→	<i>kik-u</i>
<i>sin- -Ø -ru ##</i>	→	<i>sin- -u -Ø ##</i>	→	<i>sin-u</i>	→	<i>sin-u</i>
<i>ar- -Ø -ru ##</i>	→	<i>ar- -Ø -Ø ##</i>	→	<i>ar</i>	→	<i>ari</i>

That is, the allomorphy in (50) applies in parallel with the allomorphy in (51):

(50) *-ru* → *-Ø* / *-nar* in trigrade consonant-stem verbs, multigrade vowel-stem verbs and *r*-irregular verbs

(51) *-ru* → *-Ø* / *##* in trigrade consonant-stem verbs, multigrade vowel-stem verbs and *r*-irregular verbs

This is what is traditionally called *syūsikei-setuzoku* 'conclusive-form connection'. The parallelism between (50) and (51) indicates that, before the modal suffix *-nar-*, there is a boundary which induces the same type of morphophonological computation as the main-clause-final boundary *##*. Thus, *syūsikei-setuzoku* of modal suffixes does not mean that such suffixes attach to a particular surface word form. Rather, it means that these suffixes assign to the phrase they select, namely [... *Vstem-ru*], the same

⁶ In OJ, the form *ari-nari* suggests that the surface-illicit consonant cluster *-m-* in *ar-nar* was resolved by *i*-epenthesis, as in UF *ar-Ø-ru-nar-Ø-ru* → IF *ar-nar* → SF *ari-nari*.

morphophonological computation as that induced by the main-clause-final boundary. It should be noted, however, that when the suffix *-mazi-* selects the phrase [... *ar-Ø-ru*], the allomorphy *-ru* → *-Ø*, which would yield **ar-Ø-Ø-mazi*, does not take place, as shown by the derivation *ar-Ø-ru-mazi-ku-ru* → *ar-Ø-ru-mazi-Ø-Ø* → *ar-ru-mazi* → SF *ar-u-mazi*. Before *-rasi*, by contrast, the allomorphy *-ru* → *-Ø* is optional, as indicated by the derivation UF *ar-Ø-ru-rasi* → *ar-Ø-Ø-rasi* / *ar-Ø-ru-rasi* → *ar-rasi* / *ar-ru-rasi* → SFs *ar-asi* / *ar-u-rasi*. It must therefore be assumed that, with respect to the phrase [... *ar-Ø-ru*], the relevant morphophonological computation is specified separately for each suffix. For trigrade consonant-stem verbs and multigrade vowel-stem verbs, however, the allomorphy *-ru* → *-Ø* before these modal suffixes can be stated in general terms as in (52):

- (52) *-ru* → *-Ø* / _{-*ramu*, *-rasi*, *-mer-*, *-nar-*, *-be-*, *-mazi-*} in trigrade consonant-stem verbs and multigrade vowel-stem verbs

Accordingly, the set {-*ramu*, *-rasi*, *-mer-*, *-nar-*, *-be-*, *-mazi-*} constitutes another irrealis set. This set conditions the allomorphy of the immediately preceding suffix *-ru*, and is distinct from the irrealis set {-*zu*, *-mu*, *-zi*, *-masi*, *-mahosi-*, *-yo*}, which conditions the allomorphy of the immediately preceding V thematic suffixes *-i*, *-e* and *-Ø*. The modal contents of these suffixes are heterogeneous: *-be-* marks necessity, *-mazi-* the negation of necessity, *-mer-* visual evidential modality, *-nar-* auditory evidential modality, *-ramu* non-visual conjecture about an eventuality in progress at the time of utterance and *-rasi* confident inference based on firm evidence or reasons. Their common function, however, may be understood as taking scope over a proposition constituted by *-ru* as actual-world-oriented and converting the actual-world commitment effected by *-ru* into some kind of modal judgement.

9. Conclusion

The present study has argued that the suffixal form *-re* in the EMJ realis paradigm — the so-called *izenkei* — is not an independent inflexional morpheme, but an allomorph of the suffix *-ru*, specified as [+ACTUAL-WORLD-COMMITMENT, 0PAST]. The suffix *-ru* marks a clause as [+ACT] and thereby presents the eventuality expressed by that clause as belonging to the actual world. It surfaces as *-re* when the clause marked by *-ru* is placed in a premise-setting environment. The allomorphy may therefore be formulated as in (53):

- (53) *-ru* [+ACT, 0PAST] → *-re* / PREMISE-SETTING-ENVIRONMENT: : *-Ø*, *-ba*, *-do(mo)*

In the conditional and concessive constructions, this environment is created respectively by the suffixes

-ba [+COND] and *-do(mo)* [+CONC], which set the protasis as the premise required for the eventuality expressed by the apodosis to stand in the relevant relation to the eventuality expressed by the protasis, the latter being presented in advance as belonging to the actual world. In the *=koso* type of *kakari-musubi*, the same function is performed by the phonologically null suffix $\emptyset$, specified as [+PREM]. Therefore, the particle *=koso* does not itself condition the allomorphy *-ru* $\rightarrow$ *-re*, but focalises a constituent not containing the finite predicate within an already constructed clause set as a premise. In the *=zo/=namu* type of *kakari-musubi*, by contrast, a topic–comment clause is first constructed, and a finite-predicate-containing topic constituent is nominalised by $\emptyset$ [+NOM(INAL)]. The clause is then set as a premise by a further $\emptyset$ [+PREM], after which *=zo* or *=namu* focalises the comment phrase. Although the clause is thus set as a premise, the intervening $\emptyset$ [+NOM] immediately following *-ru* blocks the local morphological relation between *-ru* and $\emptyset$ [+PREM] required to trigger the allomorphy *-ru* $\rightarrow$ *-re*. Consequently, *-ru* is not realised as *-re*.

Accordingly, *=koso* occurs with *-re*, whereas *=zo* and *=namu* occur with *-ru*, simply because the two types of *kakari-musubi* have different morphosyntactic and information-structural configurations. The particles themselves do not condition the predicate forms. Rather, each focus particle occurs in a structure that, independently of the particle, determines whether *-ru* is realised as *-re* or remains *-ru*. This reverses the traditional explanation of *kakari-musubi*, according to which the preceding focus particle would determine the form of the predicate. In this account, the morphosyntactic–semantic and information-structural configuration is constructed FIRST, and this structure THEN determines both the predicate form and which focus particle may occur. What has traditionally been regarded as direct concord between a particle and a predicate form is therefore a surface correspondence resulting from their occurrence within the same structure, constructed independently of the so-called concord.

This study further discusses the morphosyntactic parallelism between verbal and verbal-adjectival realis forms that supports the underlying presence of the same suffix *-ru* in structures such as *tab-e-ru-* and *taka-ku-ru-*. The realis forms of verbs and verbal adjectives can thus be treated as instantiations of one and the same underlying structure and one and the same process of allomorphy same allomorphic process. The analysis also accounts for the opposition between *Vstem-ru-ba* [+ACT, +COND] and *Vstem- $\emptyset$ -ba* [–ACT, +COND], and situates this opposition within the wider feature system that also includes *-sika* [+ACT, +PAST] and *-se* [–ACT, +PAST] in the environment *Vstem_-ba*. The loss of the productive opposition between *-re*_[+ACT]-*ba* and *- $\emptyset$* _[–ACT]-*ba* is proposed to have deprived *-re* of its oppositional value [+ACT]. The resulting semantic and morphological opacity allowed *-re-ba* to be reanalysed as the single conditional suffix *-reba*, specified only as [+COND] and, consequently, available for hypothetical conditions. Although this diachronic scenario requires fuller philological verification, the proposed analysis appears to provide a unified account of the morphosyntax, semantics and subsequent development of the EMJ *izenkei*.

Finally, the analysis has shown that, in terms of the four semantic natural classes identified there — namely, the realis set $\{-ru [+ACT]\}$, the irrealis set $\{-\emptyset [-ACT], -zu, -mu, -zi, -masi, -mahosi-, -yo\}$, which conditions the allomorphy of the immediately preceding V thematic suffix, the irrealis set $\{-ramu, -rasi, -mer-, -nar-, -be-, -mazi-\}$, which conditions the allomorphy of the immediately preceding suffix $-ru$, and the premise-setting set $\{-\emptyset [+PREM], -ba, -do(mo)\}$ — and the eight local allomorphic alternations in (54) = (43) plus (52), and (55) = (44), the domains traditionally treated separately — the $=koso$ type of *kakari-musubi*, conditional expressions and the irrealis–realis opposition — can be reduced to a single system:

(54) V thematic suffixes $-i$ and $-e \rightarrow -u / _ -ru$ in multigrade vowel-stem verbs

V thematic suffix $-\emptyset \rightarrow -u / _ -ru$ in trigrade consonant-stem verbs

V thematic suffix $-\emptyset \rightarrow -a / _ \{-\emptyset [-ACT], -zu, -mu, -zi, -masi, -mahosi-\}$

V thematic suffix $-i \rightarrow -e / _ \{-\emptyset [-ACT], -zu, -mu, -zi, -masi, -mahosi-, -yo\}$ in the *s*-irregular verb

V thematic suffix $-i \rightarrow -o / _ \{-\emptyset [-ACT], -zu, -mu, -zi, -masi, -mahosi-, -yo\}$ in the *k*-irregular verb

(55) $-ru \rightarrow -\emptyset / _ \#\#$ in trigrade consonant-stem verbs, multigrade vowel-stem verbs and *r*-irregular verbs

$-ru \rightarrow -\emptyset / _ \{-ramu, -rasi, -mer-, -nar-, -be-, -mazi-\}$ in trigrade consonant-stem verbs and multigrade vowel-stem verbs

$-ru \rightarrow -re / _ \{-\emptyset [+PREM], -ba, -do(mo)\}$

References

- Frellesvig, B. (2010). *A History of the Japanese Language*. Cambridge University Press.
- Hayata, T. (2000). The liquid and stem-final vowel alternations of verbs in Ancient Japanese. *Gengo Kenkyu*, 118, 5–27. https://doi.org/10.11435/gengo1939.2000.118_5
- Inoue, M. (2002). Modariti [Modality]. In T. Ōnishi (Ed.), *Hōgen bunpō chōsa gaidobukku* [Guidebook for dialect grammar surveys] (pp. 133–150). Research report for JSPS KAKENHI Grant 10410097.
- Kawamura, T., Kashiwagi, Y., & Ikura, F. (Eds.). (2023). *Kin'yō wakashū* [Collection of golden leaves of Japanese poetry]. Tokyo: Iwanami Shoten. (Iwanami Bunko, Yellow 30-1)
- Makino, S. (2025). *A New Analysis of the Morphosyntactic–Semantic Structure of Japanese Verbal Adjectives — KU-adjectives*. Preprint, LingBuzz: <https://ling.auf.net/lingbuzz/009112>; Zenodo: <https://doi.org/10.5281/zenodo.15764311>
- Nichieisha Henshūjo (Ed.). (1999). *Shin yōsetsu bungo bunpō* [New Concise Grammar of Classical Japanese] (3rd rev. ed.). Tokyo: Nichieisha.
- Oda, M. (2003). *Kodai Nihongo Bunpou* [Ancient Japanese Grammar]. Tokyo: Chikuma Shobo.
- Okimori, T. (2017). *Nihongo Zenshi* [A Comprehensive Chronological History of the Japanese Language]. Tokyo: Chikuma Shobo.
- Ōno, S., Satake, A., & Maeda, K. (Eds.). (1990). *Iwanami Kogo Jiten* [Iwanami Dictionary of Old Japanese] (revised ed.). Tokyo: Iwanami Shoten.
- Saeki, U. (Ed.). (1981). *Kokin wakashū* [Collection of Japanese poems of ancient and modern times]. Iwanami Shoten. (Iwanami Bunko, Yellow 12-1).
- Sakakura, A. (Ed.). (1970). *Taketori monogatari*. Iwanami Shoten. (Iwanami Bunko, Yellow 7-1).
- Sano, K. (2022). A compositional account of counterfactual conditional clauses in Old Japanese. *Proceedings of the Linguistic Society of America*, 7(1), Article 5280. <https://doi.org/10.3765/plsa.v7i1.5280>
- Satake, A., Yamada, H., Kudō, R., Ōtani, M., & Yamazaki, Y. (Eds.). (2013). *Man'yōshū* (Vol. 1) [Collection of ten thousand leaves]. Iwanami Shoten.
- Saussure, F. de. (1916). *Cours de linguistique générale* (C. Bally & A. Sechehaye, Eds.; avec la collaboration de A. Riedlinger). Paris: Pyot, 1971.
- Tamura, S. (2009). Tense and modality in Japanese causal expressions. In Y. Takubo, T. Kinuhata, S. Grzelak, & K. Nagai (Eds.), *Japanese/Korean linguistics* (Vol. 16, pp. 496–510). CSLI Publications.